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DTK 22.10.1
兼容性手册

DTK-221001-003-0218-222

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修订记录：

日期	修正内容
2022.10.31	创建文档
2022.11.22	更新操作系统与内核兼容性列表，补充支持的其他操作系统

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1 操作系统和内核兼容性列表

系统	版本	内核
CentOS	7.6	3.10.0-957.el7.x86_64
CentOS	8.5	4.18.0-348.el8.x86_64
Ubuntu	20.04.1	5.4.0-42-generic
NFS	3.2	3.10.0-957.nfs.5.x86_64
NFS	4.0	4.19.113-11.nfs4.x86_64
Ubuntu	22.04	5.15.0-43-generic
UOS	1021e	4.19.90-2109.1.0.0108.up2.uel20.x86_64
Kylin	v10 SP2	4.19.90-24.4.v2101.ky10.x86_64
Anolis	8.4	4.19.91-23.4.an8.x86_64
Anolis	8.6	4.19.91-26.an8.x86_64
NFS	4.0-Desktop	5.4.0-49-generic
openEuler	22.03	5.10.0-60.18.0.50.oe2203.x86_64

2 异构程序兼容性

基于 DTK-21.10、DTK-21.10.1、DTK-22.04、DTK-22.04.1、DTK-22.04.2、DTK-22.10 编译的异构可执行程序和异构共享库可以直接在 DTK-22.10.1 上使用。

3 CUDA API 兼容性列表

3.1 Device Management

CUDA	HIP	CUDA version*
cudaChooseDevice	hipChooseDevice	
cudaDeviceGetAttribute	hipDeviceGetAttribute	
cudaDeviceGetByPCIBusId	hipDeviceGetByPCIBusId	

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cudaDeviceGetCacheConfig	hipDeviceGetCacheConfig	
cudaDeviceGetLimit	hipDeviceGetLimit	
cudaDeviceGetNvSciSyncAttributes		10.2
cudaDeviceGetPCIBusId	hipDeviceGetPCIBusId	
cudaDeviceGetSharedMemConfig	hipDeviceGetSharedMemConfig	
cudaDeviceGetStreamPriorityRange	hipDeviceGetStreamPriorityRange	
cudaDeviceReset	hipDeviceReset	
cudaDeviceSetCacheConfig	hipDeviceSetCacheConfig	
cudaDeviceSetLimit		
cudaDeviceSetSharedMemConfig	hipDeviceSetSharedMemConfig	
cudaDeviceSynchronize	hipDeviceSynchronize	
cudaGetDevice	hipGetDevice	
cudaGetDeviceCount	hipGetDeviceCount	
cudaGetDeviceFlags		
cudaGetDeviceProperties	hipGetDeviceProperties	
cudaIpcCloseMemHandle	hipIpcCloseMemHandle	
cudaIpcGetEventHandle		
cudaIpcGetMemHandle	hipIpcGetMemHandle	

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cudaIpcOpenEventHandle		
cudaIpcOpenMemHandle	hipIpcOpenMemHandle	
cudaSetDevice	hipSetDevice	
cudaSetDeviceFlags	hipSetDeviceFlags	
cudaSetValidDevices		
cudaDeviceGetP2PAttribute		8.0

3.2 Error Handling

CUDA	HIP	CUDA version*
cudaGetErrorName	hipGetErrorName	
cudaGetErrorString	hipGetErrorString	
cudaGetLastError	hipGetLastError	
cudaPeekAtLastError	hipPeekAtLastError	

3.3 Stream Management

CUDA	HIP	CUDA version*
cudaStreamAddCallback	hipStreamAddCallback	
cudaStreamAttachMemAsync		
cudaStreamBeginCapture		10.0
cudaStreamEndCapture		10.0
cudaStreamIsCapturing		10.0
cudaStreamGetCaptureInfo		10.1
cudaStreamCreate	hipStreamCreate	
cudaStreamCreateWithFlags	hipStreamCreateWithFlags	
cudaStreamCreateWithPriority	hipStreamCreateWithP	

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	riority	
cudaStreamDestroy	hipStreamDestroy	
cudaStreamGetFlags	hipStreamGetFlags	
cudaStreamGetPriority	hipStreamGetPriority	
cudaStreamQuery	hipStreamQuery	
cudaStreamSynchronize	hipStreamSynchronize	
cudaStreamWaitEvent	hipStreamWaitEvent	
cudaThreadExchangeStreamCaptureMode		10.1

3.4 Event Management

CUDA	HIP	CUDA version*
cudaEventCreate	hipEventCreate	
cudaEventCreateWithFlags	hipEventCreateWithFlags	
cudaEventDestroy	hipEventDestroy	
cudaEventElapsedTime	hipEventElapsedTime	
cudaEventQuery	hipEventQuery	
cudaEventRecord	hipEventRecord	
cudaEventSynchronize	hipEventSynchronize	

3.5 External Resource Interoperability

CUDA	HIP	CUDA version*
cudaSignalExternalSemaphoresAsync		10.0
cudaWaitExternalSemaphoresAsync		10.0
cudaImportExternalMemory		10.0
cudaExternalMemoryGetMappedBuffer		10.0
cudaExternalMemoryGetMappedMipmappedArray		10.0
cudaDestroyExternalMemory		10.0
cudaImportExternalSemaphore		10.0
cudaDestroyExternalSemaphore		10.0

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3.6 Execution Control

CUDA	HIP	CUDA version*
cudaFuncGetAttributes	hipFuncGetAttributes	
cudaFuncSetAttribute		9.0
cudaFuncSetCacheConfig	hipFuncSetCacheConfig	
cudaFuncSetSharedMemConfig		
cudaGetParameterBuffer		
cudaGetParameterBufferV2		
cudaLaunchKernel	hipLaunchKernel	
cudaSetDoubleForDevice		
cudaSetDoubleForHost		
cudaLaunchCooperativeKernel		9.0
cudaLaunchCooperativeKernel MultiDevice		9.0
cudaLaunchHostFunc		10.0

3.7 Occupancy

CUDA	HIP	CUDA version*
cudaOccupancyMaxActiveBlocksPer Multiprocessor		
cudaOccupancyMaxActiveBlocksPer MultiprocessorWithFlags		

3.8 Execution Control [DEPRECATED since 7.0]

CUDA	HIP	CUDA version*
cudaConfigureCall	hipConfigureCall	
cudaLaunch	hipLaunchByPtr	

cudaSetupArgument	hipSetupArgument	

3.9 Memory Management

CUDA	HIP	CUDA version*
cudaArrayGetInfo		
cudaFree	hipFree	
cudaFreeArray	hipFreeArray	
cudaFreeHost	hipHostFree	
cudaFreeMipmappedArray		
cudaGetMipmappedArrayLevel		
cudaGetSymbolAddress	hipGetSymbolAddress	
cudaGetSymbolSize	hipGetSymbolSize	
cudaHostAlloc	hipHostMalloc	
cudaHostGetDevicePointer	hipHostGetDevicePointer	
cudaHostGetFlags	hipHostGetFlags	
cudaHostRegister	hipHostRegister	
cudaHostUnregister	hipHostUnregister	
cudaMalloc	hipMalloc	
cudaMalloc3D	hipMalloc3D	
cudaMalloc3DArray	hipMalloc3DArray	
cudaMallocArray	hipMallocArray	
cudaMallocHost	hipHostMalloc	
cudaMallocManaged	hipMallocManaged	
cudaMallocMipmappedArray		
cudaMallocPitch		
cudaMemGetInfo	hipMemGetInfo	
cudaMemPrefetchAsync		8.0

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cudaMemcpy	hipMemcpy	
cudaMemcpy2D	hipMemcpy2D	
cudaMemcpy2DArrayToArray		
cudaMemcpy2DAsync	hipMemcpy2DAsync	
cudaMemcpy2DFromArray		
cudaMemcpy2DFromArrayAsync		
cudaMemcpy2DToArray	hipMemcpy2DToArray	
cudaMemcpy2DToArrayAsync		
cudaMemcpy3D	hipMemcpy3D	
cudaMemcpy3DAsync		
cudaMemcpy3DPeer		
cudaMemcpy3DPeerAsync		
cudaMemcpyArrayToArray		
cudaMemcpyAsync	hipMemcpyAsync	
cudaMemcpyFromArray	hipMemcpyFromArray	
cudaMemcpyFromArrayAsync		
cudaMemcpyFromSymbol	hipMemcpyFromSymbol	
cudaMemcpyFromSymbolAsync	hipMemcpyFromSymbolAsync	
cudaMemcpyPeer	hipMemcpyPeer	
cudaMemcpyPeerAsync	hipMemcpyPeerAsync	
cudaMemcpyToArray	hipMemcpyToArray	
cudaMemcpyToArrayAsync		
cudaMemcpyToSymbol	hipMemcpyToSymbol	
cudaMemcpyToSymbolAsync	hipMemcpyToSymbolAsync	
cudaMemset	hipMemset	
cudaMemset2D	hipMemset2D	
cudaMemset2DAsync	hipMemset2DAsync	
cudaMemset3D	hipMemset3D	

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cudaMemset3DAsync	hipMemset3DAsync	
cudaMemsetAsync	hipMemsetAsync	
make_cudaExtent	make_hipExtent	
make_cudaPitchedPtr		
make_cudaPos	make_hipPos	

3.10 Unified Addressing

CUDA	HIP	CUDA version*
cudaPointerGetAttributes	hipPointerGetAttributes	

3.11 Peer Device Memory Access

CUDA	HIP	CUDA version*
cudaDeviceCanAccessPeer	hipDeviceCanAccessPeer	
cudaDeviceDisablePeerAccess	hipDeviceDisablePeerAccess	
cudaDeviceEnablePeerAccess	hipDeviceEnablePeerAccess	

3.12 Version Management

CUDA	HIP	CUDA version*
cudaDriverGetVersion	hipDriverGetVersion	
cudaRuntimeGetVersion	hipRuntimeGetVersion	

3.13 Graph Management

CUDA	HIP	CUDA version*
cudaGraphAddChildGraphNode		10.0
cudaGraphAddDependencies		10.0
cudaGraphAddEmptyNode		10.0
cudaGraphAddHostNode		10.0
cudaGraphAddKernelNode		10.0
cudaGraphAddMemcpyNode		10.0
cudaGraphAddMemsetNode		10.0
cudaGraphChildGraphNodeGetGraph		10.0

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cudaGraphClone		10.0
cudaGraphCreate		10.0
cudaGraphDestroy		10.0
cudaGraphDestroyNode		10.0
cudaGraphExecDestroy		10.0
cudaGraphGetEdges		10.0
cudaGraphGetNodes		10.0
cudaGraphGetRootNodes		10.0
cudaGraphHostNodeGetParams		10.0
cudaGraphHostNodeSetParams		10.0
cudaGraphInstantiate		10.0
cudaGraphExecKernelNodeSetParams		10.1
cudaGraphExecMemcpyNodeSetParams		10.2
cudaGraphExecMemsetNodeSetParams		10.2
cudaGraphExecHostNodeSetParams		10.2
cudaGraphExecUpdate		10.2
cudaGraphKernelNodeGetParams		10.0
cudaGraphKernelNodeSetParams		10.0
cudaGraphLaunch		10.0
cudaGraphMemcpyNodeGetParams		10.0
cudaGraphMemcpyNodeSetParams		10.0
cudaGraphMemsetNodeGetParams		10.0
cudaGraphMemsetNodeSetParams		10.0
cudaGraphNodeFindInClone		10.0
cudaGraphNodeGetDependencies		10.0
cudaGraphNodeGetDependentNodes		10.0
cudaGraphNodeGetType		10.0
cudaGraphRemoveDependencies		10.0

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3.14 Profiler Control

CUDA	HIP	CUDA version*
cudaProfilerInitialize		
cudaProfilerStart	hipProfilerStart	
cudaProfilerStop	hipProfilerStop	

3.15 Device Library 支持列表

数学函数支持列表

printf	umul64hi	__float2half_rn	__ull2double_rz
__popc	mul64hi	__half2float	__ull2float_rd
__popcll	__umulhi	__float2int_rd	__ull2float_rn
__clz	mulhi	__float2int_rn	__ull2float_ru
__clzll	mulhi	__float2int_ru	__ull2float_rz
__ffs	mulhi	__float2int_rz	float2int
__ffsll	__urhadd	__float2ll_rd	float2uint
__ffs	__usad	__float2ll_rn	int2float
__ffsll	__lane_id	__float2ll_ru	uint2float
__brev	__mbcnt_lo	__float2ll_rz	_clock64
__brevll	__mbcnt_hi	__float2uint_rd	_clock
__lastbit_u32_u64	__hip_ds_bpermute	__float2uint_rn	lock64
__bitextract_u32	__hip_ds_bpermutef	__float2uint_ru	lock
__bitextract_u64	__hip_ds_permute	__float2uint_rz	__named_sync
__bitinsert_u32	__hip_ds_permutef	__float2ull_rd	__clock64
__bitinsert_u64	__hip_ds_swizzle_N	__float2ull_rn	__clock

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__byte_perm	__hip_ds_swizzlef_N	__float2ull_ru	clock64
__hadd	__hip_move_dpp_N	__float2ull_rz	clock
__mul24	__shfl	__float_as_int	__named_sync
__mul64hi	__shfl_up	float_as_int	__all
__mulhi	__shfl_down	__float_as_uint	__any
__rhadd	__shfl_xor	float_as_uint	__ballot
__sad	__double2float_rd	__hiloint2double	__ballot64
__uhadd	__double2float_rn	__int2double_rn	__lanemask_gt
__umul24	__double2float_ru	__int2float_rd	__lanemask_lt
__umul64hi	__double2float_rz	__int2float_rn	__lanemask_eq
__umulhi	__double2hiint	__int2float_ru	__local_to_generic
__urhadd	__double2loint	__int2float_rz	__get_dynamicgroupbaseptr
__usad	__double2int_rd	__int_as_float	__get_dynamicgroupbaseptr
__byte_perm	__double2int_rn	int_as_float	__amdgcn_get_dynamicgroupbaseptr
__hadd	__double2int_ru	__ll2double_rd	__threadfence
__mul24	__double2int_rz	__ll2double_rn	__threadfence_block
mul24	__double2ll_rd	__ll2double_ru	__threadfence_system
__mul64hi	__double2ll_rn	__ll2double_rz	abort
mul64hi	__double2ll_ru	__ll2float_rd	_wassert

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__mulhi	__double2ll_rz	__ll2float_rn	__assert_fail
mulhi	__double2uint_rd	__ll2float_ru	__assertfail
__rhadd	__double2uint_rn	__ll2float_rz	__work_group_barrier
__sad	__double2uint_ru	__longlong_as_double	__barrier
__uhadd	__double2uint_rz	__uint2double_rn	__syncthreads
__umul24	__double2ull_rd	__uint2float_rd	__syncthreads_count
umul24	__double2ull_rn	__uint2float_rn	__syncthreads_and
__umul64hi	__double2ull_ru	__uint2float_ru	__syncthreads_or
mul64hi	__double2ull_rz	__uint2float_rz	__smid
umul64hi	__double_as_longlong	__uint_as_float	__hip_memcpy
umul64hi	__float2half_rn	__ull2double_rd	__hip_memset
mul64hi	__half2float	__ull2double_rn	memcpy
	__half	__ull2double_ru	memset

原子函数支持列表

atomicCAS	atomicMin
atomicCAS_system	atomicMin_system
atomicAdd	atomicOr
atomicAdd_system	atomicOr_system
atomicAddNoRet	atomicXor
atomicSub	atomicXor_system
atomicSub_system	atomicInc
atomicExch	atomicDec
atomicExch_system	atomicAnd

	atomicAnd_system

4 CUDA API 支持列表

HIP 系统提供了对 CUDA 函数的直接支持，以下 CUDA 函数可以直接使用。

cudaDeviceReset	cudaGetMipmappedArrayLevel
cudaDeviceSetCacheConfig	cudaMemcpy3D
cudaDeviceSetSharedMemConfig	cudaArrayGetInfo
cudaDeviceGetByPCIBusId	cudaMemcpy
cudaDeviceGetPCIBusId	cudaMemcpyPeer
cudaIpcGetEventHandle	cudaMemcpyToArray
cudaIpcOpenEventHandle	cudaMemcpyFromArray
cudaIpcGetMemHandle	cudaMemcpy2D
cudaIpcOpenMemHandle	cudaMemcpy2DToArray
cudaIpcCloseMemHandle	cudaMemcpy2DFromArray
cudaChooseDevice	cudaMemcpyToSymbol
cudaSetDevice	cudaMemcpyFromSymbol
cudaSetDeviceFlags	cudaMemcpyPeerAsync
cudaGetDeviceFlags	cudaMemcpyToSymbolAsync
cudaStreamCreate	cudaMemcpyFromSymbolAsync
cudaStreamAddCallback	cudaMemset
cudaStreamSynchronize	cudaMemset2D
cudaStreamQuery	cudaMemset3D
cudaEventCreate	cudaGetSymbolAddress
cudaEventQuery	cudaGetSymbolSize
cudaEventSynchronize	cudaMemPrefetchAsync
cudaEventElapsedTime	cudaMemAdvise
cudaLaunchKernel	cudaMemRangeGetAttribute
cudaLaunchCooperativeKernel	cudaMemRangeGetAttributes
cudaLaunchCooperativeKernelMultiDevi	cudaPointerGetAttributes

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ce	
cudaFuncSetCacheConfig	cudaDeviceCanAccessPeer
cudaConfigureCall	cudaDeviceEnablePeerAccess
cudaSetupArgument	cudaDeviceDisablePeerAccess
cudaMallocHost	cudaGetChannelDesc
cudaMallocPitch	cudaBindTexture
cudaMallocArray	cudaBindTexture2D
cudaFreeHost	cudaBindTextureToArray
cudaFreeArray	cudaBindTextureToMipmappedArray
cudaFreeMipmappedArray	cudaUnbindTexture
cudaHostAlloc	cudaGetTextureAlignmentOffset
cudaHostRegister	cudaGetTextureReference
cudaHostUnregister	cudaCreateTextureObject
cudaHostGetDevicePointer	cudaDestroyTextureObject
cudaHostGetFlags	cudaGetTextureObjectResourceDesc
cudaMalloc3D	cudaGetTextureObjectTextureDesc
cudaMalloc3DArray	cudaGetTextureObjectResourceViewDesc
cudaMallocMipmappedArray	cudaDriverGetVersion
cudaDeviceReset	cudaGetMipmappedArrayLevel

5 数学库兼容性列表

5.1 BLAS

5.2 BLAS Data types

type	CUDA	HIP	CUDA version*
define	CUBLAS_VER_MAJOR		10.1 Update 2

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define	CUBLAS_VER_MINOR		10.1 Update 2
define	CUBLAS_VER_PATCH		10.1 Update 2
define	CUBLAS_VER_BUILD		10.1 Update 2
define	CUBLAS_VERSION		10.1 Update 2
enum	cublasStatus_t	hipblasStatus_t	
0	CUBLAS_STATUS_SUCCESS	HIPBLAS_STATUS_SUCCESS	
1	CUBLAS_STATUS_NOT_INITIALIZED	HIPBLAS_STATUS_NOT_INITIALIZED	
3	CUBLAS_STATUS_ALLOC_FAILED	HIPBLAS_STATUS_ALLOC_FAILED	
7	CUBLAS_STATUS_INVALID_VALUE	HIPBLAS_STATUS_INVALID_VALUE	
8	CUBLAS_STATUS_ARCH_MISMATCH	HIPBLAS_STATUS_ARCH_MISMATCH	
11	CUBLAS_STATUS_MAPPING_ERROR	HIPBLAS_STATUS_MAPPING_ERROR	
13	CUBLAS_STATUS_EXECUTION_FAILED	HIPBLAS_STATUS_EXECUTION_FAILED	
14	CUBLAS_STATUS_INTERNAL_ERROR	HIPBLAS_STATUS_INTERNAL_ERROR	
15	CUBLAS_STATUS_NOT_SUPPORTED	HIPBLAS_STATUS_NOT_SUPPORTED	
16	CUBLAS_STATUS_LICENSE_ERROR		

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enum	cublasOperation_t	hipblasOperation_t	
0	CUBLAS_OP_N	HIPBLAS_OP_N	
1	CUBLAS_OP_T	HIPBLAS_OP_T	
2	CUBLAS_OP_C	HIPBLAS_OP_C	
2	CUBLAS_OP_HERMITAN	HIPBLAS_OP_C	10.1
3	CUBLAS_OP_CONJG		10.1
enum	cublasFillMode_t	hipblasFillMode_t	
0	CUBLAS_FILL_MODE_LOWER	HIPBLAS_FILL_MODE_LOWER	
1	CUBLAS_FILL_MODE_UPPER	HIPBLAS_FILL_MODE_UPPER	
2	CUBLAS_FILL_MODE_FULL	HIPBLAS_FILL_MODE_FULL	10.1
enum	cublasDiagType_t	hipblasDiagType_t	
0	CUBLAS_DIAG_NON_UNIT	HIPBLAS_DIAG_NON_UNIT	
1	CUBLAS_DIAG_UNIT	HIPBLAS_DIAG_UNIT	
enum	cublasSideMode_t	hipblasSideMode_t	
0	CUBLAS_SIDE_LEFT	HIPBLAS_SIDE_LEFT	
1	CUBLAS_SIDE_RIGHT	HIPBLAS_SIDE_RIGHT	
enum	cublasPointerMode_t	hipblasPointerMode_t	
0	CUBLAS_POINTER_MODE_HOST	HIPBLAS_POINTER_MODE_HOST	

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1	CUBLAS_POINTER_MODE_DEVICE	HIPBLAS_POINTER_MODE_DEVICE	
enum	cublasAtomicsMode_t		
0	CUBLAS_ATOMICS_NOT_ALLOWED		
1	CUBLAS_ATOMICS_ALLOWED		
enum	cublasGemmAlgo_t	hipblasGemmAlgo_t	8
-1	CUBLAS_GEMM_DFALT	HIPBLAS_GEMM_DEFAULT	8
-1	CUBLAS_GEMM_DEFAULT	HIPBLAS_GEMM_DEFAULT	8
0	CUBLAS_GEMM_ALGO0		8
1	CUBLAS_GEMM_ALGO1		8
2	CUBLAS_GEMM_ALGO2		8
3	CUBLAS_GEMM_ALGO3		8
4	CUBLAS_GEMM_ALGO4		8
5	CUBLAS_GEMM_ALGO5		8
6	CUBLAS_GEMM_ALGO6		8
7	CUBLAS_GEMM_ALGO7		8
8	CUBLAS_GEMM_AL		9

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	GO8		
9	CUBLAS_GEMM_ALGO9		9
10	CUBLAS_GEMM_ALGO10		9
11	CUBLAS_GEMM_ALGO11		9
12	CUBLAS_GEMM_ALGO12		9
13	CUBLAS_GEMM_ALGO13		9
14	CUBLAS_GEMM_ALGO14		9
15	CUBLAS_GEMM_ALGO15		9
16	CUBLAS_GEMM_ALGO16		9
17	CUBLAS_GEMM_ALGO17		9
18	CUBLAS_GEMM_ALGO18		9.2
19	CUBLAS_GEMM_ALGO19		9.2
20	CUBLAS_GEMM_ALGO20		9.2
21	CUBLAS_GEMM_ALGO21		9.2
22	CUBLAS_GEMM_ALGO22		9.2

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23	CUBLAS_GEMM_ALGO23		9.2
99	CUBLAS_GEMM_DEFAULT_TENSOR_OP		9
99	CUBLAS_GEMM_DFALT_TENSOR_OP		9
100	CUBLAS_GEMM_ALGO0_TENSOR_OP		9
101	CUBLAS_GEMM_ALGO1_TENSOR_OP		9
102	CUBLAS_GEMM_ALGO2_TENSOR_OP		9
103	CUBLAS_GEMM_ALGO3_TENSOR_OP		9
104	CUBLAS_GEMM_ALGO4_TENSOR_OP		9
105	CUBLAS_GEMM_ALGO5_TENSOR_OP		9.2
106	CUBLAS_GEMM_ALGO6_TENSOR_OP		9.2
107	CUBLAS_GEMM_ALGO7_TENSOR_OP		9.2
108	CUBLAS_GEMM_ALGO8_TENSOR_OP		9.2
109	CUBLAS_GEMM_ALGO9_TENSOR_OP		9.2
110	CUBLAS_GEMM_ALGO10_TENSOR_OP		9.2
111	CUBLAS_GEMM_AL		9.2

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	GO11_TENSOR_OP		
112	CUBLAS_GEMM_ALGO12_TENSOR_OP		9.2
113	CUBLAS_GEMM_ALGO13_TENSOR_OP		9.2
114	CUBLAS_GEMM_ALGO14_TENSOR_OP		9.2
115	CUBLAS_GEMM_ALGO15_TENSOR_OP		9.2
enum	cublasMath_t		9
0	CUBLAS_DEFAULT_MATH		9
1	CUBLAS_TENSOR_OP_MATH		9
enum*	cublasDataType_t		7.5
struct	cublasContext		
struct*	cublasHandle_t	hipblasHandle_t	

5.2.1 BLAS Level-1 Function Reference

CUDA	HIP	CUDA version*
cublasCaxpy	hipblasCaxpy	
cublasCaxpy_v2	hipblasCaxpy	
cublasCcopy	hipblasCcopy	
cublasCcopy_v2	hipblasCcopy	
cublasCdotc	hipblasCdotc	
cublasCdotc_v2	hipblasCdotc	
cublasCdotu	hipblasCdotu	
cublasCdotu_v2	hipblasCdotu	
cublasCrot	hipblasCrot	

cublasCrot_v2	hipblasCrot	
cublasCrotg	hipblasCrotg	
cublasCrotg_v2	hipblasCrotg	
cublasCscal	hipblasCscal	
cublasCscal_v2	hipblasCscal	
cublasCsrot	hipblasCsrot	
cublasCsrot_v2	hipblasCsrot	
cublasCsscal	hipblasCsscal	
cublasCsscal_v2	hipblasCsscal	
cublasCswap	hipblasCswap	
cublasCswap_v2	hipblasCswap	
cublasDasum	hipblasDasum	
cublasDasum_v2	hipblasDasum	
cublasDaxpy	hipblasDaxpy	
cublasDaxpy_v2	hipblasDaxpy	
cublasDcopy	hipblasDcopy	
cublasDcopy_v2	hipblasDcopy	
cublasDdot	hipblasDdot	
cublasDdot_v2	hipblasDdot	
cublasDnrm2	hipblasDnrm2	
cublasDnrm2_v2	hipblasDnrm2	
cublasDrot	hipblasDrot	
cublasDrot_v2	hipblasDrot	
cublasDrotg	hipblasDrotg	
cublasDrotg_v2	hipblasDrotg	
cublasDrotm	hipblasDrotm	
cublasDrotm_v2	hipblasDrotm	
cublasDrotmg	hipblasDrotmg	

cublasDrotmg_v2	hipblasDrotmg	
cublasDscal	hipblasDscal	
cublasDscal_v2	hipblasDscal	
cublasDswap	hipblasDswap	
cublasDswap_v2	hipblasDswap	
cublasDzasum	hipblasDzasum	
cublasDzasum_v2	hipblasDzasum	
cublasDznrm2	hipblasDznrm2	
cublasDznrm2_v2	hipblasDznrm2	
cublasIcamax	hipblasIcamax	
cublasIcamax_v2	hipblasIcamax	
cublasIcamin	hipblasIcamin	
cublasIcamin_v2	hipblasIcamin	
cublasIdamax	hipblasIdamax	
cublasIdamax_v2	hipblasIdamax	
cublasIdamin	hipblasIdamin	
cublasIdamin_v2	hipblasIdamin	
cublasIsamax	hipblasIsamax	
cublasIsamax_v2	hipblasIsamax	
cublasIsamin	hipblasIsamin	
cublasIsamin_v2	hipblasIsamin	
cublasIzamax	hipblasIzamax	
cublasIzamax_v2	hipblasIzamax	
cublasIzamin	hipblasIzamin	
cublasIzamin_v2	hipblasIzamin	
cublasNrm2Ex	hipblasNrm2Ex	8
cublasSasum	hipblasSasum	
cublasSasum_v2	hipblasSasum	

cublasSaxpy	hipblasSaxpy	
cublasSaxpy_v2	hipblasSaxpy	
cublasScasum	hipblasScasum	
cublasScasum_v2	hipblasScasum	
cublasScnrm2	hipblasScnrm2	
cublasScnrm2_v2	hipblasScnrm2	
cublasScopy	hipblasScopy	
cublasScopy_v2	hipblasScopy	
cublasSdot	hipblasSdot	
cublasSdot_v2	hipblasSdot	
cublasSnrm2	hipblasSnrm2	
cublasSnrm2_v2	hipblasSnrm2	
cublasSrot	hipblasSrot	
cublasSrot_v2	hipblasSrot	
cublasSrotg	hipblasSrotg	
cublasSrotg_v2	hipblasSrotg	
cublasSrotm	hipblasSrotm	
cublasSrotm_v2	hipblasSrotm	
cublasSrotmg	hipblasSrotmg	
cublasSrotmg_v2	hipblasSrotmg	
cublasSscal	hipblasSscal	
cublasSscal_v2	hipblasSscal	
cublasSswap	hipblasSswap	
cublasSswap_v2	hipblasSswap	
cublasZaxpy	hipblasZaxpy	
cublasZaxpy_v2	hipblasZaxpy	
cublasZcopy	hipblasZcopy	
cublasZcopy_v2	hipblasZcopy	

cublasZdotc	hipblasZdotc	
cublasZdotc_v2	hipblasZdotc	
cublasZdotu	hipblasZdotu	
cublasZdotu_v2	hipblasZdotu	
cublasZdrot	hipblasZdrot	
cublasZdrot_v2	hipblasZdrot	
cublasZdscal	hipblasZdscal	
cublasZdscal_v2	hipblasZdscal	
cublasZrot	hipblasZrot	
cublasZrot_v2	hipblasZrot	
cublasZrotg	hipblasZrotg	
cublasZrotg_v2	hipblasZrotg	
cublasZscal	hipblasZscal	
cublasZscal_v2	hipblasZscal	
cublasZswap	hipblasZswap	
cublasZswap_v2	hipblasZswap	

5.2.2 BLAS Level-2 Function Reference

CUDA	HIP	CUDA version*
cublasCgbmv	hipblasCgbmv	
cublasCgbmv_v2	hipblasCgbmv	
cublasCgemv	hipblasCgemv	
cublasCgemv_v2	hipblasCgemv	
cublasCgerc	hipblasCgerc	
cublasCgerc_v2	hipblasCgerc	
cublasCgeru	hipblasCgeru	
cublasCgeru_v2	hipblasCgeru	
cublasChbmv	hipblasChbmv	
cublasChbmv_v2	hipblasChbmv	

cublasChemv	hipblasChemv	
cublasChemv_v2	hipblasChemv	
cublasCher	hipblasCher	
cublasCher2	hipblasCher2	
cublasCher2_v2	hipblasCher2	
cublasCher_v2	hipblasCher	
cublasChpmv	hipblasChpmv	
cublasChpmv_v2	hipblasChpmv	
cublasChpr	hipblasChpr	
cublasChpr2	hipblasChpr2	
cublasChpr2_v2	hipblasChpr2	
cublasChpr_v2	hipblasChpr	
cublasCsymv	hipblasCsymv	
cublasCsymv_v2	hipblasCsymv	
cublasCsyr	hipblasCsyr	
cublasCsyr2	hipblasCsyr2	
cublasCsyr2_v2	hipblasCsyr2	
cublasCsyr_v2	hipblasCsyr	
cublasCtbmv	hipblasCtbmv	
cublasCtbmv_v2	hipblasCtbmv	
cublasCtbsv	hipblasCtbsv	
cublasCtbsv_v2	hipblasCtbsv	
cublasCtpmv	hipblasCtpmv	
cublasCtpmv_v2	hipblasCtpmv	
cublasCtpsv	hipblasCtpsv	
cublasCtpsv_v2	hipblasCtpsv	
cublasCtrmv	hipblasCtrmv	
cublasCtrmv_v2	hipblasCtrmv	

cublasCtrsv	hipblasCtrsv	
cublasCtrsv_v2	hipblasCtrsv	
cublasDgbmv	hipblasDgbmv	
cublasDgbmv_v2	hipblasDgbmv	
cublasDgemv	hipblasDgemv	
cublasDgemv_v2	hipblasDgemv	
cublasDger	hipblasDger	
cublasDger_v2	hipblasDger	
cublasDsbmv	hipblasDsbmv	
cublasDsbmv_v2	hipblasDsbmv	
cublasDspmv	hipblasDspmv	
cublasDspmv_v2	hipblasDspmv	
cublasDspr	hipblasDspr	
cublasDspr2	hipblasDspr2	
cublasDspr2_v2	hipblasDspr2	
cublasDspr_v2	hipblasDspr	
cublasDsymv	hipblasDsymv	
cublasDsymv_v2	hipblasDsymv	
cublasDsyr	hipblasDsyr	
cublasDsyr2	hipblasDsyr2	
cublasDsyr2_v2	hipblasDsyr2	
cublasDsyr_v2	hipblasDsyr	
cublasDtbmv	hipblasDtbmv	
cublasDtbmv_v2	hipblasDtbmv	
cublasDtbsv	hipblasDtbsv	
cublasDtbsv_v2	hipblasDtbsv	
cublasDtpmv	hipblasDtpmv	
cublasDtpmv_v2	hipblasDtpmv	

cublasDtpsv	hipblasDtpsv	
cublasDtpsv_v2	hipblasDtpsv	
cublasDtrmv	hipblasDtrmv	
cublasDtrmv_v2	hipblasDtrmv	
cublasDtrsv	hipblasDtrsv	
cublasDtrsv_v2	hipblasDtrsv	
cublasSgbmv	hipblasSgbmv	
cublasSgbmv_v2	hipblasSgbmv	
cublasSgemv	hipblasSgemv	
cublasSgemv_v2	hipblasSgemv	
cublasSger	hipblasSger	
cublasSger_v2	hipblasSger	
cublasSsbmv	hipblasSsbmv	
cublasSsbmv_v2	hipblasSsbmv	
cublasSspmv	hipblasSspmv	
cublasSspmv_v2	hipblasSspmv	
cublasSspr	hipblasSspr	
cublasSspr2	hipblasSspr2	
cublasSspr2_v2	hipblasSspr2	
cublasSspr_v2	hipblasSspr	
cublasSsymv	hipblasSsymv	
cublasSsymv_v2	hipblasSsymv	
cublasSsyr	hipblasSsyr	
cublasSsyr2	hipblasSsyr2	
cublasSsyr2_v2	hipblasSsyr2	
cublasSsyr_v2	hipblasSsyr	
cublasStbmv	hipblasStbmv	
cublasStbmv_v2	hipblasStbmv	

cublasStbsv	hipblasStbsv	
cublasStbsv_v2	hipblasStbsv	
cublasStpmv	hipblasStpmv	
cublasStpmv_v2	hipblasStpmv	
cublasStpsv	hipblasStpsv	
cublasStpsv_v2	hipblasStpsv	
cublasStrmv	hipblasStrmv	
cublasStrmv_v2	hipblasStrmv	
cublasStrsv	hipblasStrsv	
cublasStrsv_v2	hipblasStrsv	
cublasZgbmv	hipblasZgbmv	
cublasZgbmv_v2	hipblasZgbmv	
cublasZgemv	hipblasZgemv	
cublasZgemv_v2	hipblasZgemv	
cublasZgerc	hipblasZgerc	
cublasZgerc_v2	hipblasZgerc	
cublasZgeru	hipblasZgeru	
cublasZgeru_v2	hipblasZgeru	
cublasZhbmv	hipblasZhbmv	
cublasZhbmv_v2	hipblasZhbmv	
cublasZhemv	hipblasZhemv	
cublasZhemv_v2	hipblasZhemv	
cublasZher	hipblasZher	
cublasZher2	hipblasZher2	
cublasZher2_v2	hipblasZher2	
cublasZher_v2	hipblasZher	
cublasZhpmv	hipblasZhpmv	
cublasZhpmv_v2	hipblasZhpmv	

cublasZhpr	hipblasZhpr	
cublasZhpr2	hipblasZhpr2	
cublasZhpr2_v2	hipblasZhpr2	
cublasZhpr_v2	hipblasZhpr	
cublasZsymv	hipblasZsymv	
cublasZsymv_v2	hipblasZsymv	
cublasZsyr	hipblasZsyr	
cublasZsyr2	hipblasZsyr2	
cublasZsyr2_v2	hipblasZsyr2	
cublasZsyr_v2	hipblasZsyr	
cublasZtbmv	hipblasZtbmv	
cublasZtbmv_v2	hipblasZtbmv	
cublasZtbsv	hipblasZtbsv	
cublasZtbsv_v2	hipblasZtbsv	
cublasZtpmv	hipblasZtpmv	
cublasZtpmv_v2	hipblasZtpmv	
cublasZtpsv	hipblasZtpsv	
cublasZtpsv_v2	hipblasZtpsv	
cublasZtrmv	hipblasZtrmv	
cublasZtrmv_v2	hipblasZtrmv	
cublasZtrsv	hipblasZtrsv	
cublasZtrsv_v2	hipblasZtrsv	

5.2.3 BLAS Level-3 Function Reference

CUDA	HIP	CUDA version*
cublasCgemm	hipblasCgemm	
cublasCgemm3m		8
cublasCgemm3mBatched		8
cublasCgemm3mEx		8

cublasCgemm3mStrided Batched		8
cublasCgemmBatched	hipblasCgemmBatched	
cublasCgemmStridedBat ched	hipblasCgemmStridedBatc hed	8
cublasCgemm_v2	hipblasCgemm	
cublasChemm	hipblasChemm	
cublasChemm_v2	hipblasChemm	
cublasCher2k	hipblasCher2k	
cublasCher2k_v2	hipblasCher2k	
cublasCherk	hipblasCherk	
cublasCherk_v2	hipblasCherk	
cublasCherkx	hipblasCherkx	
cublasCsymm	hipblasCsymm	
cublasCsymm_v2	hipblasCsymm	
cublasCsyr2k	hipblasCsyr2k	
cublasCsyr2k_v2	hipblasCsyr2k	
cublasCsyrk	hipblasCsyrk	
cublasCsyrk_v2	hipblasCsyrk	
cublasCsyrkx	hipblasCsyrkx	
cublasCtrmm	hipblasCtrmm	
cublasCtrmm_v2	hipblasCtrmm	
cublasCtrsm	hipblasCtrsm	
cublasCtrsm_v2	hipblasCtrsm	
cublasDgemm	hipblasDgemm	
cublasDgemmBatched	hipblasDgemmBatched	
cublasDgemmStridedBat ched	hipblasDgemmStridedBatc hed	8

cublasDgemm_v2	hipblasDgemm	
cublasDsymm	hipblasDsymm	
cublasDsymm_v2	hipblasDsymm	
cublasDsyr2k	hipblasDsyr2k	
cublasDsyr2k_v2	hipblasDsyr2k	
cublasDsyrk	hipblasDsyrk	
cublasDsyrk_v2	hipblasDsyrk	
cublasDsyrkx	hipblasDsyrkx	
cublasDtrmm	hipblasDtrmm	
cublasDtrmm_v2	hipblasDtrmm	
cublasDtrsm	hipblasDtrsm	
cublasDtrsm_v2	hipblasDtrsm	
cublasHgemm	hipblasHgemm	7.5
cublasHgemmBatched	hipblasHgemmBatched	9
cublasHgemmStridedBatched	hipblasHgemmStridedBatched	8
cublasSgemm	hipblasSgemm	
cublasSgemmBatched	hipblasSgemmBatched	
cublasSgemmStridedBatched	hipblasSgemmStridedBatched	8
cublasSgemm_v2	hipblasSgemm	
cublasSsymm	hipblasSsymm	
cublasSsymm_v2	hipblasSsymm	
cublasSsyr2k	hipblasSsyr2k	
cublasSsyr2k_v2	hipblasSsyr2k	
cublasSsyrk	hipblasSsyrk	
cublasSsyrk_v2	hipblasSsyrk	
cublasSsyrkx	hipblasSsyrkx	

cublasStrmm	hipblasStrmm	
cublasStrmm_v2	hipblasStrmm	
cublasStrsm	hipblasStrsm	
cublasStrsm_v2	hipblasStrsm	
cublasZgemm	hipblasZgemm	
cublasZgemm3m		8
cublasZgemmBatched	hipblasZgemmBatched	
cublasZgemmStridedBatched	hipblasZgemmStridedBatched	8
cublasZgemm_v2	hipblasZgemm	
cublasZhemm	hipblasZhemm	
cublasZhemm_v2	hipblasZhemm	
cublasZher2k	hipblasZher2k	
cublasZher2k_v2	hipblasZher2k	
cublasZherk	hipblasZherk	
cublasZherk_v2	hipblasZherk	
cublasZherkx	hipblasZherkx	
cublasZsymm	hipblasZsymm	
cublasZsymm_v2	hipblasZsymm	
cublasZsyr2k	hipblasZsyr2k	
cublasZsyr2k_v2	hipblasZsyr2k	
cublasZsyrk	hipblasZsyrk	
cublasZsyrk_v2	hipblasZsyrk	
cublasZsyrkx	hipblasZsyrkx	
cublasZtrmm	hipblasZtrmm	
cublasZtrmm_v2	hipblasZtrmm	
cublasZtrsm	hipblasZtrsm	
cublasZtrsm_v2	hipblasZtrsm	

5.2.4 BLAS Helper Function Referenc

CUDA	HIP	CUDA version*
cublasAlloc		
cublasCreate	hipblasCreate	
cublasCreate_v2	hipblasCreate	
cublasDestroy	hipblasDestroy	
cublasDestroy_v2	hipblasDestroy	
cublasFree		
cublasGetAtomicsMode	hipblasGetAtomicsMode	
cublasGetCudartVersion		10.1
cublasGetError		
cublasGetLoggerCallback		9.2
cublasGetMathMode		9
cublasGetMatrix	hipblasGetMatrix	
cublasGetMatrixAsync	hipblasGetMatrixAsync	
cublasGetPointerMode	hipblasGetPointerMode	
cublasGetPointerMode_v2	hipblasGetPointerMode	
cublasGetProperty		
cublasGetSmCountTarget		11.3
cublasGetStatusName		11.4
cublasGetStatusString		11.4
cublasGetStream	hipblasGetStream	
cublasGetStream_v2	hipblasGetStream	
cublasGetVector	hipblasGetVector	
cublasGetVectorAsync	hipblasGetVectorAsync	
cublasGetVersion		

cublasGetVersion_v2		
cublasInit		
cublasLogCallback		9.2
cublasLoggerConfigure		9.2
cublasMigrateComputeType		11
cublasSetAtomicsMode	hipblasSetAtomicsMode	
cublasSetKernelStream		
cublasSetLoggerCallback		9.2
cublasSetMathMode		
cublasSetMatrix	hipblasSetMatrix	
cublasSetMatrixAsync	hipblasSetMatrixAsync	
cublasSetPointerMode	hipblasSetPointerMode	
cublasSetPointerMode_v2	hipblasSetPointerMode	
cublasSetSmCountTarget		11.3
cublasSetStream	hipblasSetStream	
cublasSetStream_v2	hipblasSetStream	
cublasSetVector	hipblasSetVector	
cublasSetVectorAsync	hipblasSetVectorAsync	
cublasShutdown		
cublasXerbla		

5.2.5 BLAS-like Extension

CUDA	HIP	CUDA version*
cublasAsumEx		10.1
cublasAxpyEx	hipblasAxpyEx	8
cublasCdgmm	hipblasCdgmm	
cublasCgeam	hipblasCgeam	

cublasCgelsBatched		
cublasCgemmEx		8
cublasCgeqrfBatched	hipblasCgeqrfBatched	
cublasCgetrfBatched	hipblasCgetrfBatched	
cublasCgetriBatched	hipblasCgetriBatched	
cublasCgetrsBatched	hipblasCgetrsBatched	
cublasCherk3mEx		8
cublasCherkEx		8
cublasCmatinvBatched		
cublasCopyEx		10.1
cublasCsyrk3mEx		8
cublasCsyrkEx		8
cublasCtpttr		
cublasCtrsmBatched	hipblasCtrsmBatched	
cublasCtrttp		
cublasDdgmm	hipblasDdgmm	
cublasDgeam	hipblasDgeam	
cublasDgelsBatched		
cublasDgeqrfBatched	hipblasDgeqrfBatched	
cublasDgetrfBatched	hipblasDgetrfBatched	
cublasDgetriBatched	hipblasDgetriBatched	
cublasDgetrsBatched	hipblasDgetrsBatched	
cublasDmatinvBatched		
cublasDotEx	hipblasDotEx	8
cublasDotcEx	hipblasDotcEx	8
cublasDtpttr		
cublasDtrsmBatched	hipblasDtrsmBatched	
cublasDtrttp		

cublasGemmBatchedEx	hipblasGemmBatchedEx	9.1
cublasGemmEx	hipblasGemmEx	8
cublasGemmStridedBatchedEx	hipblasGemmStridedBatchedEx	9.1
cublasIamaxEx		10.1
cublasIaminEx		10.1
cublasRotEx	hipblasRotEx	10.1
cublasRotgEx		10.1
cublasRotmEx		10.1
cublasRotmgEx		10.1
cublasScalEx	hipblasScalEx	8
cublasSdgmm	hipblasSdgmm	
cublasSgeam	hipblasSgeam	
cublasSgelsBatched		
cublasSgemmEx		7.5
cublasSgeqrfBatched	hipblasSgeqrfBatched	
cublasSgetrfBatched	hipblasSgetrfBatched	
cublasSgetriBatched	hipblasSgetriBatched	
cublasSgetrsBatched	hipblasSgetrsBatched	
cublasSmatinvBatched		
cublasStpttr		
cublasStrsmBatched	hipblasStrsmBatched	
cublasStrttp		
cublasSwapEx		10.1
cublasUint8gemmBias		8
cublasZdgmm	hipblasZdgmm	
cublasZgeam	hipblasZgeam	
cublasZgelsBatched		

cublasZgeqrfBatched	hipblasZgeqrfBatched	
cublasZgetrfBatched	hipblasZgetrfBatched	
cublasZgetriBatched	hipblasZgetriBatched	
cublasZgetrsBatched	hipblasZgetrsBatched	
cublasZmatinvBatched		
cublasZtpttr		
cublasZtrsmBatched	hipblasZtrsmBatched	
cublasZtrttp		

5.3 MIOpen

CUDA	HIP	CUDA version *
cudnnActivationBackward	hipdnnActivationBackward	
cudnnActivationForward	hipdnnActivationForward	
cudnnAddTensor	hipdnnAddTensor	
cudnnAdvInferVersionCheck		8.0.1
cudnnAdvTrainVersionCheck		8.0.1
cudnnBackendCreateDescriptor		8.0.1
cudnnBackendDestroyDescriptor		8.0.1
cudnnBackendExecute		8.0.1
cudnnBackendFinalize		8.0.1
cudnnBackendGetAttribute		8.0.1
cudnnBackendInitialize		8.0.1
cudnnBackendSetAttribute		8.0.1
cudnnBatchNormalizationBackward	hipdnnBatchNormalizationBackward	
cudnnBatchNormalizationBackwardEx		7.4.1

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cudnnBatchNormalizationForwardInference	hipdnnBatchNormalizationForwardInference	
cudnnBatchNormalizationForwardTraining	hipdnnBatchNormalizationForwardTraining	
cudnnBatchNormalizationForwardTrainingEx		7.4.1
cudnnBuildRNNDynamic		8.0.1
cudnnCTCLoss		7.0.5
cudnnCTCLoss_v8		8.0.1
cudnnCnnInferVersionCheck		8.0.2
cudnnCnnTrainVersionCheck		8.0.2
cudnnConvolutionBackwardBias	hipdnnConvolutionBackwardBias	
cudnnConvolutionBackwardData	hipdnnConvolutionBackwardData	
cudnnConvolutionBackwardFilter	hipdnnConvolutionBackwardFilter	
cudnnConvolutionBiasActivationForward		6.0.0
cudnnConvolutionForward	hipdnnConvolutionForward	
cudnnCreate	hipdnnCreate	
cudnnCreateActivationDescriptor	hipdnnCreateActivationDescriptor	
cudnnCreateAttnDescriptor		7.5.0
cudnnCreateCTCLossDescriptor		7.0.5
cudnnCreateConvolutionDescriptor	hipdnnCreateConvolutionDescriptor	
cudnnCreateDropoutDescriptor	hipdnnCreateDropoutDescrip	

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	tor	
cudnnCreateFilterDescriptor	hipdnnCreateFilterDescriptor	
cudnnCreateFusedOpsConstParamPack		7.6.0
cudnnCreateFusedOpsPlan		7.6.0
cudnnCreateFusedOpsVariantParamPack		7.6.0
cudnnCreateLRNDescriptor	hipdnnCreateLRNDescriptor	
cudnnCreateOpTensorDescriptor	hipdnnCreateOpTensorDescriptor	
cudnnCreatePoolingDescriptor	hipdnnCreatePoolingDescriptor	
cudnnCreateRNNDataDescriptor		7.2.1
cudnnCreateRNNDescriptor	hipdnnCreateRNNDescriptor	
cudnnCreateReduceTensorDescriptor	hipdnnCreateReduceTensorDescriptor	
cudnnCreateSeqDataDescriptor		7.5.0
cudnnCreateSpatialTransformerDescriptor		5.0.0
cudnnCreateTensorDescriptor	hipdnnCreateTensorDescriptor	
cudnnCreateTensorTransformDescriptor		7.5.0
cudnnDeriveBNTensorDescriptor	hipdnnDeriveBNTensorDescriptor	
cudnnDeriveNormTensorDescriptor		8.0.1
cudnnDestroy	hipdnnDestroy	
cudnnDestroyActivationDescriptor	hipdnnDestroyActivationDes	

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	criptor	
cudnnDestroyAttnDescriptor		7.5.0
cudnnDestroyCTCLossDescriptor		7.0.5
cudnnDestroyConvolutionDescriptor	hipdnnDestroyConvolutionDescriptor	
cudnnDestroyDropoutDescriptor	hipdnnDestroyDropoutDescriptor	
cudnnDestroyFilterDescriptor	hipdnnDestroyFilterDescriptor	
cudnnDestroyFusedOpsConstParamPack		7.6.0
cudnnDestroyFusedOpsPlan		7.6.0
cudnnDestroyFusedOpsVariantParamPack		7.6.0
cudnnDestroyLRNDescriptor	hipdnnDestroyLRNDescriptor	
cudnnDestroyOpTensorDescriptor	hipdnnDestroyOpTensorDescriptor	
cudnnDestroyPoolingDescriptor	hipdnnDestroyPoolingDescriptor	
cudnnDestroyRNNDataDescriptor		7.2.1
cudnnDestroyRNNDescriptor	hipdnnDestroyRNNDescriptor	
cudnnDestroyReduceTensorDescriptor	hipdnnDestroyReduceTensorDescriptor	
cudnnDestroySeqDataDescriptor		7.5.0
cudnnDestroySpatialTransformerDescriptor		5.0.0

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cudnnDestroyTensorDescriptor	hipdnnDestroyTensorDescriptor	
cudnnDestroyTensorTransformDescriptor		7.5.0
cudnnDivisiveNormalizationBackward		3.0.0
cudnnDivisiveNormalizationForward		3.0.0
cudnnDropoutBackward		5.0.0
cudnnDropoutForward		5.0.0
cudnnDropoutGetReserveSpaceSize		5.0.0
cudnnDropoutGetStatesSize	hipdnnDropoutGetStatesSize	
cudnnFindConvolutionBackwardDataAlgorithm	hipdnnFindConvolutionBackwardDataAlgorithm	
cudnnFindConvolutionBackwardDataAlgorithmEx	hipdnnFindConvolutionBackwardDataAlgorithmEx	
cudnnFindConvolutionBackwardFilterAlgorithm	hipdnnFindConvolutionBackwardFilterAlgorithm	
cudnnFindConvolutionBackwardFilterAlgorithmEx	hipdnnFindConvolutionBackwardFilterAlgorithmEx	
cudnnFindConvolutionForwardAlgorithm	hipdnnFindConvolutionForwardAlgorithm	
cudnnFindConvolutionForwardAlgorithmEx	hipdnnFindConvolutionForwardAlgorithmEx	
cudnnFusedOpsExecute		7.6.0
cudnnGetActivationDescriptor	hipdnnGetActivationDescriptor	
cudnnGetActivationDescriptorSwishBeta		8.2.0

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cudnnGetAttnDescriptor		7.5.0
cudnnGetBatchNormalizationBackwardExWorkspaceSize		7.4.1
cudnnGetBatchNormalizationForwardTrainingExWorkspaceSize		7.4.1
cudnnGetBatchNormalizationTrainingExReserveSpaceSize		7.4.1
cudnnGetCTCLossDescriptor		7.0.5
cudnnGetCTCLossDescriptorEx		7.5.0
cudnnGetCTCLossDescriptor_v8		8.0.1
cudnnGetCTCLossWorkspaceSize		7.0.5
cudnnGetCTCLossWorkspaceSize_v8		8.0.1
cudnnGetCallback		7.1.3
cudnnGetConvolution2dDescriptor	hipdnnGetConvolution2dDescriptor	
cudnnGetConvolution2dForwardOutputDim	hipdnnGetConvolution2dForwardOutputDim	
cudnnGetConvolutionBackwardDataAlgorithmMaxCount		7.0.5
cudnnGetConvolutionBackwardDataAlgorithm_v7		7.0.5
cudnnGetConvolutionBackwardDataWorkspaceSize	hipdnnGetConvolutionBackwardDataWorkspaceSize	
cudnnGetConvolutionBackwardFilterAlgorithmMaxCount		7.0.5
cudnnGetConvolutionBackwardFilterAlgorithm_v7		7.0.5
cudnnGetConvolutionBackwardFilter	hipdnnGetConvolutionBackw	

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WorkspaceSize	ardFilterWorkspaceSize	
cudnnGetConvolutionForwardAlgorithmMaxCount		7.0.5
cudnnGetConvolutionForwardAlgorithm_v7		7.0.5
cudnnGetConvolutionForwardWorkspaceSize	hipdnnGetConvolutionForwardWorkspaceSize	
cudnnGetConvolutionGroupCount		7.0.5
cudnnGetConvolutionMathType		7.0.5
cudnnGetConvolutionNdDescriptor		2.0.0
cudnnGetConvolutionNdForwardOutputDim		2.0.0
cudnnGetConvolutionReorderType		7.6.0
cudnnGetCudartVersion		6.0.0
cudnnGetDropoutDescriptor		7.0.5
cudnnGetErrorString	hipdnnGetErrorString	
cudnnGetFilter4dDescriptor	hipdnnGetFilter4dDescriptor	
cudnnGetFilterNdDescriptor	hipdnnGetFilterNdDescriptor	
cudnnGetFilterSizeInBytes		7.6.0
cudnnGetFoldedConvBackwardDataDescriptors		7.6.0
cudnnGetFusedOpsConstParamPackAttribute		7.6.0
cudnnGetFusedOpsVariantParamPackAttribute		7.6.0
cudnnGetLRNDescriptor	hipdnnGetLRNDescriptor	
cudnnGetMultiHeadAttnBuffers		7.5.0
cudnnGetMultiHeadAttnWeights		7.5.0

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cudnnGetNormalizationBackwardWorkspaceSize		8.0.1
cudnnGetNormalizationForwardTrainingWorkspaceSize		8.0.1
cudnnGetNormalizationTrainingReserveSpaceSize		8.0.1
cudnnGetOpTensorDescriptor	hipdnnGetOpTensorDescriptor	
cudnnGetPooling2dDescriptor	hipdnnGetPooling2dDescriptor	
cudnnGetPooling2dForwardOutputDim	hipdnnGetPooling2dForwardOutputDim	
cudnnGetPoolingNdDescriptor		2.0.0
cudnnGetPoolingNdForwardOutputDim		2.0.0
cudnnGetProperty		6.0.0
cudnnGetRNNDataDescriptor		7.2.1
cudnnGetRNNDescriptor_v8		8.0.1
cudnnGetRNNTempSpaceSizes		8.0.1
cudnnGetRNNTrainingReserveSize	hipdnnGetRNNTrainingReserveSize	
cudnnGetRNNWeightParams		8.0.1
cudnnGetRNNWeightSpaceSize		8.0.1
cudnnGetRNNWorkspaceSize	hipdnnGetRNNWorkspaceSize	
cudnnGetReduceTensorDescriptor	hipdnnGetReduceTensorDescriptor	
cudnnGetReductionIndicesSize		6.0.0

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cudnnGetReductionWorkspaceSize	hipdnnGetReductionWorkspaceSize	
cudnnGetSeqDataDescriptor		7.5.0
cudnnGetStream	hipdnnGetStream	
cudnnGetTensor4dDescriptor	hipdnnGetTensor4dDescriptor	
cudnnGetTensorNdDescriptor	hipdnnGetTensorNdDescriptor	
cudnnGetTensorSizeInBytes		6.0.0
cudnnGetTensorTransformDescriptor		7.5.0
cudnnGetVersion	hipdnnGetVersion	
cudnnIm2Col		2.0.0
cudnnInitTransformDest		7.5.0
cudnnLRNCrossChannelBackward	hipdnnLRNCrossChannelBackward	
cudnnLRNCrossChannelForward	hipdnnLRNCrossChannelForward	
cudnnMakeFusedOpsPlan		7.6.0
cudnnMultiHeadAttnBackwardData		7.5.0
cudnnMultiHeadAttnBackwardWeights		7.5.0
cudnnMultiHeadAttnForward		7.5.0
cudnnNormalizationBackward		8.0.1
cudnnNormalizationForwardInference		8.0.1
cudnnNormalizationForwardTraining		8.0.1
cudnnOpTensor	hipdnnOpTensor	
cudnnOpsInferVersionCheck		8.0.1
cudnnOpsTrainVersionCheck		8.0.1

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cudnnPoolingBackward	hipdnnPoolingBackward	
cudnnPoolingForward	hipdnnPoolingForward	
cudnnQueryRuntimeError		7.0.5
cudnnRNNBackwardData	hipdnnRNNBackwardData	
cudnnRNNBackwardData_v8		8.0.2
cudnnRNNBackwardWeights	hipdnnRNNBackwardWeights	
cudnnRNNBackwardWeights_v8		8.0.2
cudnnRNNForward		8.0.1
cudnnRNNForwardInference	hipdnnRNNForwardInference	
cudnnRNNForwardTraining	hipdnnRNNForwardTraining	
cudnnRNNGetClip_v8		8.0.1
cudnnRNNSetClip_v8		8.0.1
cudnnReduceTensor	hipdnnReduceTensor	
cudnnReorderFilterAndBias		7.6.0
cudnnRestoreDropoutDescriptor		7.0.5
cudnnScaleTensor	hipdnnScaleTensor	
cudnnSetActivationDescriptor	hipdnnSetActivationDescriptor	
cudnnSetActivationDescriptorSwishBeta		8.2.0
cudnnSetAttnDescriptor		7.5.0
cudnnSetCTCLossDescriptor		7.0.5
cudnnSetCTCLossDescriptorEx		7.6.0
cudnnSetCTCLossDescriptor_v8		8.0.1
cudnnSetCallback		7.1.3
cudnnSetConvolution2dDescriptor	hipdnnSetConvolution2dDescriptor	

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cudnnSetConvolutionGroupCount	hipdnnSetConvolutionGroupCount	
cudnnSetConvolutionMathType	hipdnnSetConvolutionMathType	
cudnnSetConvolutionNdDescriptor	hipdnnSetConvolutionNdDescriptor	
cudnnSetConvolutionReorderType		7.6.0
cudnnSetDropoutDescriptor	hipdnnSetDropoutDescriptor	
cudnnSetFilter4dDescriptor	hipdnnSetFilter4dDescriptor	
cudnnSetFilterNdDescriptor	hipdnnSetFilterNdDescriptor	
cudnnSetFusedOpsConstParamPackAttribute		7.6.0
cudnnSetFusedOpsVariantParamPackAttribute		7.6.0
cudnnSetLRNDescriptor	hipdnnSetLRNDescriptor	
cudnnSetOpTensorDescriptor	hipdnnSetOpTensorDescriptor	
cudnnSetPooling2dDescriptor	hipdnnSetPooling2dDescriptor	
cudnnSetPoolingNdDescriptor	hipdnnSetPoolingNdDescriptor	
cudnnSetRNNDataDescriptor		7.2.1
cudnnSetRNNDescriptor_v8		8.0.1
cudnnSetReduceTensorDescriptor	hipdnnSetReduceTensorDescriptor	
cudnnSetSeqDataDescriptor		7.5.0
cudnnSetSpatialTransformerNdDescriptor		5.0.0

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cudnnSetStream	hipdnnSetStream	
cudnnSetTensor	hipdnnSetTensor	
cudnnSetTensor4dDescriptor	hipdnnSetTensor4dDescriptor	
cudnnSetTensor4dDescriptorEx	hipdnnSetTensor4dDescriptorEx	
cudnnSetTensorNdDescriptor	hipdnnSetTensorNdDescriptor	
cudnnSetTensorNdDescriptorEx		6.0.0
cudnnSetTensorTransformDescriptor		7.5.0
cudnnSoftmaxBackward	hipdnnSoftmaxBackward	
cudnnSoftmaxForward	hipdnnSoftmaxForward	
cudnnSpatialTfGridGeneratorBackward		5.0.0
cudnnSpatialTfGridGeneratorForward		5.0.0
cudnnSpatialTfSamplerBackward		5.0.0
cudnnSpatialTfSamplerForward		5.0.0
cudnnTransformFilter		7.6.0
cudnnTransformTensor		2.0.0
cudnnTransformTensorEx		7.5.0

5.4 FFT

5.4.1 Data types

CUDA	HIP	CUDAversion*
CUFFT_ALLOC_FAILED	HIPFFT_ALLOC_FAILED	
CUFFT_C2C	HIPFFT_C2C	
CUFFT_C2R	HIPFFT_C2R	
CUFFT_CB_LD_COMPLEX	HIPFFT_CB_LD_COMPLEX	
CUFFT_CB_LD_COMPLEX_DOUBLE	HIPFFT_CB_LD_COMPLEX_DOUBLE	

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CUFFT_CB_LD_REAL	HIPFFT_CB_LD_REAL	
CUFFT_CB_LD_REAL_DOUBLE	HIPFFT_CB_LD_REAL_DOUBLE	
CUFFT_CB_ST_COMPLEX	HIPFFT_CB_ST_COMPLEX	
CUFFT_CB_ST_COMPLEX_DOUBLE	HIPFFT_CB_ST_COMPLEX_DOUBLE	
CUFFT_CB_ST_REAL	HIPFFT_CB_ST_REAL	
CUFFT_CB_ST_REAL_DOUBLE	HIPFFT_CB_ST_REAL_DOUBLE	
CUFFT_CB_UNDEFINED	HIPFFT_CB_UNDEFINED	
CUFFT_COMPATIBILITY_DEFAULT		
CUFFT_COMPATIBILITY_FFTW_PADDING		
CUFFT_COPY_DEVICE_TO_DEVICE		
CUFFT_COPY_DEVICE_TO_HOST		
CUFFT_COPY_HOST_TO_DEVICE		
CUFFT_COPY_UNDEFINED		
CUFFT_D2Z	HIPFFT_D2Z	
CUFFT_EXEC_FAILED	HIPFFT_EXEC_FAILED	
CUFFT_FORMAT_UNDEFINED		
CUFFT_FORWARD	HIPFFT_FORWARD	
CUFFT_INCOMPLETE_PARAMETER_LIST	HIPFFT_INCOMPLETE_PARAMETER_LIST	

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CUFFT_INTERNAL_ERROR	HIPFFT_INTERNAL_ERROR	
CUFFT_INVALID_DEVICE	HIPFFT_INVALID_DEVICE	
CUFFT_INVALID_PLAN	HIPFFT_INVALID_PLAN	
CUFFT_INVALID_SIZE	HIPFFT_INVALID_SIZE	
CUFFT_INVALID_TYPE	HIPFFT_INVALID_TYPE	
CUFFT_INVALID_VALUE	HIPFFT_INVALID_VALUE	
CUFFT_INVERSE	HIPFFT_BACKWARD	
CUFFT_LICENSE_ERROR		
CUFFT_NOT_IMPLEMENTED	HIPFFT_NOT_IMPLEMENTED	
CUFFT_NOT_SUPPORTED	HIPFFT_NOT_SUPPORTED	8
CUFFT_NO_WORKSPACE	HIPFFT_NO_WORKSPACE	
CUFFT_PARSE_ERROR	HIPFFT_PARSE_ERROR	
CUFFT_QUERY_1D_FACTORS		
CUFFT_QUERY_UNDEFINED		
CUFFT_R2C	HIPFFT_R2C	
CUFFT_SETUP_FAILED	HIPFFT_SETUP_FAILED	
CUFFT_SUCCESS	HIPFFT_SUCCESS	
CUFFT_UNALIGNED_DATA	HIPFFT_UNALIGNED_DATA	
CUFFT_VERSION		10.2
CUFFT_VER_BUILD		10.2
CUFFT_VER_MAJOR		10.2
CUFFT_VER_MINOR		10.2
CUFFT_VER_PATCH		10.2

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CUFFT_WORKAREA_MINIMAL		9.2
CUFFT_WORKAREA_PERFORMANCE		
CUFFT_WORKAREA_USER		9.2
CUFFT_XT_FORMAT_1D_INPUT_SHUFFLED		
CUFFT_XT_FORMAT_INPLACE		
CUFFT_XT_FORMAT_INPLACE_SHUFFLED		
CUFFT_XT_FORMAT_INPUT		
CUFFT_XT_FORMAT_OUTPUT		
CUFFT_Z2D	HIPFFT_Z2D	
CUFFT_Z2Z	HIPFFT_Z2Z	
MAX_CUFFT_ERROR		
cufftCompatibility		
cufftCompatibility_t		
cufftComplex	hipfftComplex	
cufftDoubleComplex	hipfftDoubleComplex	
cufftDoubleReal	hipfftDoubleReal	
cufftHandle	hipfftHandle	
cufftReal	hipfftReal	
cufftResult	hipfftResult	
cufftResult_t	hipfftResult_t	
cufftType	hipfftType	
cufftType_t	hipfftType_t	

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cufftXt1dFactors		
cufftXt1dFactors_t		
cufftXtCallbackType	hipfftXtCallbackType	
cufftXtCallbackType_t	hipfftXtCallbackType_t	
cufftXtCopyType		
cufftXtCopyType_t		
cufftXtQueryType		
cufftXtQueryType_t		
cufftXtSubFormat		
cufftXtSubFormat_t		
cufftXtWorkAreaPolicy		9.2
cufftXtWorkAreaPolicy_t		9.2

5.4.2 APIfunctions

CUDA	HIP	CUDAversion*
cufftCallbackLoadC	hipfftCallbackLoadC	
cufftCallbackLoadD	hipfftCallbackLoadD	
cufftCallbackLoadR	hipfftCallbackLoadR	
cufftCallbackLoadZ	hipfftCallbackLoadZ	
cufftCallbackStoreC	hipfftCallbackStoreC	
cufftCallbackStoreD	hipfftCallbackStoreD	
cufftCallbackStoreR	hipfftCallbackStoreR	
cufftCallbackStoreZ	hipfftCallbackStoreZ	
cufftCreate	hipfftCreate	
cufftDestroy	hipfftDestroy	
cufftEstimate1d	hipfftEstimate1d	
cufftEstimate2d	hipfftEstimate2d	
cufftEstimate3d	hipfftEstimate3d	

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cufftEstimateMany	hipfftEstimateMany	
cufftExecC2C	hipfftExecC2C	
cufftExecC2R	hipfftExecC2R	
cufftExecD2Z	hipfftExecD2Z	
cufftExecR2C	hipfftExecR2C	
cufftExecZ2D	hipfftExecZ2D	
cufftExecZ2Z	hipfftExecZ2Z	
cufftGetProperty	hipfftGetProperty	8
cufftGetSize	hipfftGetSize	
cufftGetSize1d	hipfftGetSize1d	
cufftGetSize2d	hipfftGetSize2d	
cufftGetSize3d	hipfftGetSize3d	
cufftGetSizeMany	hipfftGetSizeMany	
cufftGetSizeMany64	hipfftGetSizeMany64	7.5
cufftGetVersion	hipfftGetVersion	
cufftMakePlan1d	hipfftMakePlan1d	
cufftMakePlan2d	hipfftMakePlan2d	
cufftMakePlan3d	hipfftMakePlan3d	
cufftMakePlanMany	hipfftMakePlanMany	
cufftMakePlanMany64	hipfftMakePlanMany64	7.5
cufftPlan1d	hipfftPlan1d	
cufftPlan2d	hipfftPlan2d	
cufftPlan3d	hipfftPlan3d	
cufftPlanMany	hipfftPlanMany	
cufftSetAutoAllocation	hipfftSetAutoAllocation	
cufftSetStream	hipfftSetStream	
cufftSetWorkArea	hipfftSetWorkArea	
cufftXtClearCallback	hipfftXtClearCallback	

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cufftXtExec		8
cufftXtExecDescriptor		
cufftXtExecDescriptorC2C		
cufftXtExecDescriptorC2R		
cufftXtExecDescriptorD2Z		
cufftXtExecDescriptorR2C		
cufftXtExecDescriptorZ2D		
cufftXtExecDescriptorZ2Z		
cufftXtFree		
cufftXtGetSizeMany		8
cufftXtMakePlanMany		8
cufftXtMalloc		
cufftXtMemcpy		
cufftXtQueryPlan		
cufftXtSetCallback	hipfftXtSetCallback	
cufftXtSetCallbackSharedSize	hipfftXtSetCallbackSharedSize	
cufftXtSetGPUs		
cufftXtSetWorkArea		
cufftXtSetWorkAreaPolicy		9.2

5.5 SOLVER

CUDA	HIP	CUDA version*
cusolverDnCreate		
cusolverDnDestroy		
cusolverDnSetStream		
cusolverDnGetStream		
cusolverDnCreateSyevjInfo		

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cusolverDnDestroySyevjInfo		
cusolverDnXsyevjSetTolerance		
cusolverDnXsyevjSetMaxSweeps		
cusolverDnXsyevjSetSortEig		
cusolverDnXsyevjGetResidual		
cusolverDnXsyevjGetSweeps		
cusolverDnCreateGesvdjInfo		
cusolverDnDestroyGesvdjInfo		
cusolverDnXgesvdjSetTolerance		
cusolverDnXgesvdjSetMaxSweeps		
cusolverDnXgesvdjSetSortEig		
cusolverDnXgesvdjGetResidual		
cusolverDnXgesvdjGetSweeps		
cusolverDnIRSParamsCreate		
cusolverDnIRSParamsDestroy		
cusolverDnIRSParamsSetSolverPrecisions		
cusolverDnIRSParamsSetSolverMainPrecision		
cusolverDnIRSParamsSetSolverLowestPrecision		
cusolverDnIRSParamsSetRefinementSolver		
cusolverDnIRSParamsSetTol		
cusolverDnIRSParamsSetTolInner		
cusolverDnIRSParamsSetMaxIters		
cusolverDnIRSParamsSetMaxItersInner		
cusolverDnIRSParamsEnableFallback		
cusolverDnIRSParamsDisableFallback		
cusolverDnIRSParamsGetMaxIters		

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cusolverDnIRSInfosCreate		
cusolverDnIRSInfosDestroy		7.5
cusolverDnIRSInfosGetMaxIters		
cusolverDnIRSInfosGetNiters		
cusolverDnIRSInfosGetOuterNiters		
cusolverDnIRSInfosRequestResidual		
cusolverDnIRSInfosGetResidualHistory		
cusolverDnCreateParams		
cusolverDnDestroyParams		
cusolverDnSetAdvOptions		
cusolverDnSpotrf_bufferSize		8
cusolverDnDpotrf_bufferSize		
cusolverDnCpotrf_bufferSize		
cusolverDnZpotrf_bufferSize		
cusolverDnSpotrf	rocsolver_spotrf	
cusolverDnDpotrf	rocsolver_dpotrf	
cusolverDnCpotrf	rocsolver_cpotrf	
cusolverDnZpotrf	rocsolver_zpotrf	
cusolverDnSpotrs		
cusolverDnDpotrs		
cusolverDnCpotrs		
cusolverDnZpotrs		
cusolverDnSpotri_bufferSize		
cusolverDnDpotri_bufferSize		
cusolverDnCpotri_bufferSize		
cusolverDnZpotri_bufferSize		
cusolverDnSpotri		
cusolverDnDpotri		

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cusolverDnCpotri		
cusolverDnZpotri		
cusolverDnSgetrf_bufferSize		
cusolverDnDgetrf_bufferSize		
cusolverDnCgetrf_bufferSize		
cusolverDnZgetrf_bufferSize		
cusolverDnSgetrf	rocsolver_sgetrf	
cusolverDnDgetrf	rocsolver_dgetrf	
cusolverDnCgetrf	rocsolver_cgetrf	
cusolverDnZgetrf	rocsolver_zgetrf	
cusolverDnSgetrs	rocsolver_sgetrs	
cusolverDnDgetrs	rocsolver_dgetrs	
cusolverDnCgetrs	rocsolver_cgetrs	
cusolverDnZgetrs	rocsolver_zgetrs	
cusolverDnZZgesv_bufferSize		
cusolverDnZCgesv_bufferSize		
cusolverDnZKgesv_bufferSize		
cusolverDnZEgesv_bufferSize		
cusolverDnZYgesv_bufferSize		
cusolverDnCCgesv_bufferSize		
cusolverDnCKgesv_bufferSize		
cusolverDnCEgesv_bufferSize		
cusolverDnCYgesv_bufferSize		
cusolverDnDDgesv_bufferSize		
cusolverDnDSgesv_bufferSize		
cusolverDnDHgesv_bufferSize		
cusolverDnDBgesv_bufferSize		
cusolverDnDXgesv_bufferSize		

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cusolverDnSSgesv_bufferSize		
cusolverDnSHgesv_bufferSize		
cusolverDnSBgesv_bufferSize		
cusolverDnSXgesv_bufferSize		
cusolverDnZZgesv		
cusolverDnZCgesv		
cusolverDnZKgesv		
cusolverDnZEgesv		
cusolverDnZYgesv		
cusolverDnCCgesv		
cusolverDnCKgesv		
cusolverDnCEgesv		
cusolverDnCYgesv		
cusolverDnDDgesv		
cusolverDnDSgesv		
cusolverDnDHgesv		
cusolverDnDBgesv		
cusolverDnDXgesv		
cusolverDnSSgesv		
cusolverDnSHgesv		
cusolverDnSBgesv		9
cusolverDnSXgesv		
cusolverDnIRSXgesv_bufferSize		
cusolverDnIRSXgesv		
cusolverDnSgeqrf_bufferSize		
cusolverDnDgeqrf_bufferSize		
cusolverDnCgeqrf_bufferSize		
cusolverDnZgeqrf_bufferSize		

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cusolverDnSgeqrf	rocsolver_sgeqrf	
cusolverDnDgeqrf	rocsolver_dgeqrf	
cusolverDnCgeqrf	rocsolver_cgeqrf	
cusolverDnZgeqrf	rocsolver_zgeqrf	
cusolverDnZZgels_bufferSize		
cusolverDnZCgels_bufferSize		
cusolverDnZKgels_bufferSize		
cusolverDnZEgels_bufferSize		
cusolverDnZYgels_bufferSize		
cusolverDnCCgels_bufferSize		
cusolverDnCKgels_bufferSize		
cusolverDnCEgels_bufferSize		
cusolverDnCYgels_bufferSize		
cusolverDnDDgels_bufferSize		
cusolverDnDSgels_bufferSize		
cusolverDnDHgels_bufferSize		
cusolverDnDBgels_bufferSize		
cusolverDnDXgels_bufferSize		
cusolverDnSSgels_bufferSize		
cusolverDnSHgels_bufferSize		
cusolverDnSBgels_bufferSize		
cusolverDnSXgels_bufferSize		
cusolverDnZZgels	rocsolver_zgels	
cusolverDnZCgels		
cusolverDnZKgels		
cusolverDnZEgels		
cusolverDnZYgels		
cusolverDnCCgels	rocsolver_cgels	

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cusolverDnCKgels		
cusolverDnCEgels		
cusolverDnCYgels		
cusolverDnDDgels	rocsolver_dgels	
cusolverDnDSgels		
cusolverDnDHgels		
cusolverDnDBgels		
cusolverDnDXgels		
cusolverDnSSgels	rocsolver_sgels	
cusolverDnSHgels		
cusolverDnSBgels		
cusolverDnSXgels		
cusolverDnIRSXgels_bufferSize		
cusolverDnIRSXgels		
cusolverDnSormqr_bufferSize		
cusolverDnDormqr_bufferSize		
cusolverDnCunmqr_bufferSize		
cusolverDnZunmqr_bufferSize		
cusolverDnSormqr	rocsolver_sormqr	
cusolverDnDormqr	rocsolver_dormqr	
cusolverDnCunmqr	rocsolver_cunmqr	
cusolverDnZunmqr	rocsolver_zunmqr	
cusolverDnSorgqr_bufferSize		
cusolverDnDorgqr_bufferSize		
cusolverDnCungqr_bufferSize		
cusolverDnZungqr_bufferSize		
cusolverDnSorgqr	rocsolver_sorgqr	
cusolverDnDorgqr	rocsolver_dorgqr	

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cusolverDnCungqr	rocsolver_cungqr	
cusolverDnZungqr	rocsolver_zungqr	
cusolverDnSsytrf_bufferSize		
cusolverDnDsytrf_bufferSize		
cusolverDnCsytrf_bufferSize		
cusolverDnZsytrf_bufferSize		
cusolverDnSsytrf		
cusolverDnDsytrf		
cusolverDnCsytrf		
cusolverDnZsytrf		
cusolverDnSpotrfBatched		
cusolverDnDpotrfBatched		
cusolverDnCpotrfBatched		
cusolverDnZpotrfBatched		
cusolverDnSpotrsBatched		
cusolverDnDpotrsBatched		
cusolverDnCpotrsBatched		
cusolverDnZpotrsBatched		
cusolverDnSgebrd_bufferSize		
cusolverDnDgebrd_bufferSize		
cusolverDnCgebrd_bufferSize		
cusolverDnZgebrd_bufferSize		
cusolverDnSgebrd	rocsolver_sgebrd	
cusolverDnDgebr	rocsolver_dgebrd	
cusolverDnCgebrd	rocsolver_cgebrd	
cusolverDnZgebrd	rocsolver_zgebrd	
cusolverDnSorgbr_bufferSize		
cusolverDnDorgbr_bufferSize		

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cusolverDnCungbr_bufferSize		
cusolverDnZungbr_bufferSize		
cusolverDnSorgbr	rocsolver_sorgbr	
cusolverDnDorgbr	rocsolver_dorgbr	
cusolverDnCungbr	rocsolver_cungbr	
cusolverDnZungbr	rocsolver_zungbr	
cusolverDnSsytrd_bufferSize		
cusolverDnDsytrd_bufferSize		
cusolverDnChetrd_bufferSize		
cusolverDnZhetrd_bufferSize		
cusolverDnSsytrd	rocsolver_ssytrd	
cusolverDnDsytrd	rocsolver_dsytrd	
cusolverDnChetrd	rocsolver_chetrd	
cusolverDnZhetrd	rocsolver_zhetrd	
cusolverDnSormtr_bufferSize		
cusolverDnDormtr_bufferSize		
cusolverDnCunmtr_bufferSize		
cusolverDnZunmtr_bufferSize		
cusolverDnSormtr	rocsolver_sormtr	
cusolverDnDormtr	rocsolver_dormtr	
cusolverDnCunmtr	rocsolver_cunmtr	
cusolverDnZunmtr	rocsolver_zunmtr	
cusolverDnSorgtr_bufferSiz		
cusolverDnDorgtr_bufferSize		
cusolverDnCungtr_bufferSize		
cusolverDnZungtr_bufferSize		
cusolverDnSorgtr	rocsolver_sorgtr	
cusolverDnDorgtr	rocsolver_dorgtr	

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cusolverDnCungtr	rocsolver_cungtr	
cusolverDnZungtr	rocsolver_zungtr	
cusolverDnSgesvd_bufferSize		
cusolverDnDgesvd_bufferSize		
cusolverDnCgesvd_bufferSize		
cusolverDnZgesvd_bufferSize		
cusolverDnSgesvd	rocsolver_sgesvd	
cusolverDnDgesvd	rocsolver_dgesvd	
cusolverDnCgesvd	rocsolver_cgesvd	
cusolverDnZgesvd	rocsolver_zgesvd	
cusolverDnSgesvdj_bufferSize		
cusolverDnDgesvdj_bufferSize		
cusolverDnCgesvdj_bufferSize		
cusolverDnZgesvdj_bufferSize		
cusolverDnSgesvdj		
cusolverDnDgesvdj		
cusolverDnCgesvdj		
cusolverDnZgesvdj		
cusolverDnSgesvdjBatched_bufferSize		
cusolverDnDgesvdjBatched_bufferSize		
cusolverDnCgesvdjBatched_bufferSize		
cusolverDnZgesvdjBatched_bufferSize		
cusolverDnSgesvdjBatched		
cusolverDnDgesvdjBatched		
cusolverDnCgesvdjBatched		
cusolverDnZgesvdjBatched		
cusolverDnSgesvdaStridedBatched_bufferSize		

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cusolverDnDgesvdaStridedBatched_bufferSize		
cusolverDnCgesvdaStridedBatched_bufferSize		
cusolverDnZgesvdaStridedBatched_bufferSize		
cusolverDnSgesvdaStridedBatched		
cusolverDnDgesvdaStridedBatched		
cusolverDnCgesvdaStridedBatched		
cusolverDnZgesvdaStridedBatched		
cusolverDnSsyevd_bufferSize		
cusolverDnDsyevd_bufferSize		
cusolverDnCheevd_bufferSize		
cusolverDnZheevd_bufferSize		
cusolverDnSsyevd	rocsolver_ssyev	
cusolverDnDsyevd	rocsolver_dsyev	
cusolverDnCheevd	rocsolver_cheev	
cusolverDnZheevd	rocsolver_zheev	
cusolverDnSsyevdx_bufferSize		
cusolverDnDsyevdx_bufferSize		
cusolverDnCheevdx_bufferSize		
cusolverDnZheevdx_bufferSize		
cusolverDnSsyevdx		
cusolverDnDsyevdx		
cusolverDnCheevdx		
cusolverDnZheevdx		
cusolverDnSsygvd_bufferSize		
cusolverDnDsygvd_bufferSize		

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cusolverDnChegvd_bufferSize		
cusolverDnZhegvd_bufferSize		
cusolverDnSsygvd	rocsolver_ssygv	
cusolverDnDsygvd	rocsolver_dsygv	
cusolverDnChegvd	rocsolver_chegv	
cusolverDnZhegvd	rocsolver_zhegv	
cusolverDnSsygvdx_bufferSize		
cusolverDnDsygvdx_bufferSize		
cusolverDnChegvdx_bufferSize		
cusolverDnZhegvdx_bufferSize		
cusolverDnSsygvdx		
cusolverDnDsygvdx		
cusolverDnChegvdx		
cusolverDnZhegvdx		
cusolverDnSsyevj_bufferSize		
cusolverDnDsyevj_bufferSize		
cusolverDnCheevj_bufferSize		
cusolverDnZheevj_bufferSize		
cusolverDnSsyevj		
cusolverDnDsyevj		
cusolverDnCheevj		
cusolverDnZheevj		
cusolverDnSsygvj_bufferSize		
cusolverDnDsygvj_bufferSize		
cusolverDnChegvj_bufferSize		
cusolverDnZhegvj_bufferSize		
cusolverDnSsygvj		
cusolverDnDsygvj		

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cusolverDnChegvj		
cusolverDnZhegvj		
cusolverDnSsyevjBatched_bufferSize		
cusolverDnDsyevjBatched_bufferSize		
cusolverDnCheevjBatched_bufferSize		
cusolverDnZheevjBatched_bufferSize		
cusolverDnSsyevjBatched		
cusolverDnDsyevjBatched		
cusolverDnCheevjBatched		
cusolverDnZheevjBatched		
cusolverDnXpotrf_bufferSize		
cusolverDnXpotrf		
cusolverDnXpotrs		
cusolverDnXgetrf_bufferSize		
cusolverDnXgetrf		
cusolverDnXgetrs		
cusolverDnXgeqrf_bufferSize		
cusolverDnXgeqrf		
cusolverDnXsytrs_bufferSize		
cusolverDnXsytrs		
cusolverDnXtrtri_bufferSize		
cusolverDnXtrtri		
cusolverDnXgesvd_bufferSize		
cusolverDnXgesvd		
cusolverDnXgesvdp_bufferSize		
cusolverDnXgesvdp		
cusolverDnXgesvdr_bufferSize		
cusolverDnXgesvdr		

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cusolverDnXsyevd_bufferSize		
cusolverDnXsyevd		
cusolverDnXsyevdx_bufferSize		
cusolverDnXsyevdx		
cusolverSpCreate		
cusolverSpDestroy		
cusolverSpSetStream		
cusolverSpXcsrissymHost		
cusolverSpScsrlsvlu		
cusolverSpDcsrlsvlu		
cusolverSpCcsrlsvlu		
cusolverSpZcsrlsvlu		
cusolverSpScsrlsvqr		
cusolverSpDcsrlsvqr		
cusolverSpCcsrlsvqr		
cusolverSpZcsrlsvqr		
cusolverSpScsrlsvchol		
cusolverSpDcsrlsvchol		
cusolverSpCcsrlsvchol		
cusolverSpZcsrlsvchol		
cusolverSpScsrlsqvqr		
cusolverSpDcsrlsqvqr		
cusolverSpCcsrlsqvqr		
cusolverSpZcsrlsqvqr		
cusolverSpScsreigvsi		
cusolverSpDcsreigvsi		
cusolverSpCcsreigvsi		
cusolverSpZcsreigvsi		

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solverspScsreigs		
cusolverSpDcsreigs		
cusolverSpCcsreigs		
cusolverSpZcsreigs		
cusolverSpXcsrsymrcmHost		
cusolverSpXcsrsymmdqHost		
cusolverSpXcsrsymamdHost		
cusolverSpXcsrmetisndHost		
cusolverSpScsrzfdHost		
cusolverSpDcsrzfdHost		
cusolverSpCcsrzfdHost		
cusolverSpZcsrzfdHost		
cusolverSpXcsrperm_bufferSizeHost		
cusolverSpXcsrpermHost		
cusolverSpCreateCsrqrInfo		
cusolverSpDestroyCsrqrInfo		
cusolverSpXcsrqrAnalysisBatched		
cusolverSpScsrqrBufferInfoBatched		
cusolverSpDcsrqrBufferInfoBatched		
cusolverSpCcsrqrBufferInfoBatched		
cusolverSpZcsrqrBufferInfoBatched		
cusolverSpScsrqrsvBatched		
cusolverSpDcsrqrsvBatched		
cusolverSpCcsrqrsvBatched		
cusolverSpZcsrqrsvBatched		
cusolverRfAccessBundledFactors		
cusolverRfAnalyze		
cusolverRfSetupDevice		

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cusolverRfSetupHost		
cusolverRfCreate		
cusolverRfExtractBundledFactorsHost		
cusolverRfExtractSplitFactorsHost		
cusolverRfDestroy		
cusolverRfGetMatrixFormat		
cusolverRfGetNumericProperties		
cusolverRfGetNumericBoostReport		
cusolverRfGetResetValuesFastMode		
cusolverRfGet_Algs		
cusolverRfRefactor		
cusolverRfResetValues		
cusolverRfSetMatrixFormat		
cusolverRfSetNumericProperties		
cusolverRfSetResetValuesFastMode		
cusolverRfSetAlgs		
cusolverRfSolve		
cusolverRfBatchSetupHost		
cusolverRfBatchAnalyze		
cusolverRfBatchResetValues		
cusolverRfBatchRefactor		
cusolverRfBatchSolve		
cusolverRfBatchZeroPivot		

5.6 rocSPARSE

5.6.1 rocSPARSE 数据类型支持

type	CUDA	HIP	CUDAversion*
define	CUSPARSE_VER _MAJOR		10.1Update2

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define	CUSPARSE_VER_MINOR		10.1Update2
define	CUSPARSE_VER_PATCH		10.1Update2
define	CUSPARSE_VER_BUILD		10.1Update2
define	CUSPARSE_VERSION		10.1Update2
enum	cusparseAction_t	hipsparseAction_t	
0	CUSPARSE_ACTION_SYMBOLIC	HIPSPARSE_ACTION_SYMBOLIC	
1	CUSPARSE_ACTION_NUMERIC	HIPSPARSE_ACTION_NUMERIC	
enum	cusparseDirection_t	hipsparseDirection_t	
0	CUSPARSE_DIRECTION_ROW	HIPSPARSE_DIRECTION_ROW	
1	CUSPARSE_DIRECTION_COLUMN	HIPSPARSE_DIRECTION_COLUMN	
enum	cusparseHybPartition_t	hipsparseHybPartition_t	
0	CUSPARSE_HYB_PARTITION_AUTO	HIPSPARSE_HYB_PARTITION_AUTO	
1	CUSPARSE_HYB_PARTITION_USER	HIPSPARSE_HYB_PARTITION_USER	

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2	CUSPARSE_HYB_PARTITION_MAX	HIPSPARSE_HYB_PARTITION_MAX	
enum	cusparseDiagType_t	hipsparseDiagType_t	
0	CUSPARSE_DIAG_TYPE_NON_UNIT	HIPSPARSE_DIAG_TYPE_NON_UNIT	
1	CUSPARSE_DIAG_TYPE_UNIT	HIPSPARSE_DIAG_TYPE_UNIT	
enum	cusparseFillMode_t	hipsparseFillMode_t	
0	CUSPARSE_FILL_MODE_LOWER	HIPSPARSE_FILL_MODE_LOWER	
1	CUSPARSE_FILL_MODE_UPPER	HIPSPARSE_FILL_MODE_UPPER	
enum	cusparseIndexBase_t	hipsparseIndexBase_t	
0	CUSPARSE_INDEX_BASE_ZERO	HIPSPARSE_INDEX_BASE_ZERO	
1	CUSPARSE_INDEX_BASE_ONE	HIPSPARSE_INDEX_BASE_ONE	
enum	cusparseMatrixType_t	hipsparseMatrixType_t	
0	CUSPARSE_MATRIX_TYPE_GENERAL	HIPSPARSE_MATRIX_TYPE_GENERAL	

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1	CUSPARSE_MATRIX_TYPE_SYMMETRIC	HIPSPARSE_MATRIX_TYPE_SYMMETRIC	
2	CUSPARSE_MATRIX_TYPE_HERMITIAN	HIPSPARSE_MATRIX_TYPE_HERMITIAN	
3	CUSPARSE_MATRIX_TYPE_TRIANGULAR	HIPSPARSE_MATRIX_TYPE_TRIANGULAR	
enum	cusparseOperation_t	hipsparseOperation_t	
0	CUSPARSE_OPERATION_NON_TRANSPOSE	HIPSPARSE_OPERATION_NON_TRANSPOSE	
1	CUSPARSE_OPERATION_TRANSPOSE	HIPSPARSE_OPERATION_TRANSPOSE	
2	CUSPARSE_OPERATION_CONJUGATE_TRANSPOSE	HIPSPARSE_OPERATION_CONJUGATE_TRANSPOSE	
enum	cusparsePointerMode_t	hipsparsePointerMode_t	
0	CUSPARSE_POINTER_MODE_HOST	HIPSPARSE_POINTER_MODE_HOST	
1	CUSPARSE_POINTER_MODE_DEVICE	HIPSPARSE_POINTER_MODE_DEVICE	

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enum	cusparseAlgMode_t		8.0
0	CUSPARSE_ALG0		8.0
1	CUSPARSE_ALG1		8.0
0	CUSPARSE_ALG_NAIVE		9.2
1	CUSPARSE_ALG_MERGE_PATH		9.2
enum	cusparseSolvePolicy_t	hipsparseSolvePolicy_t	
0	CUSPARSE_SOLVE_POLICY_NO_LEVEL	HIPSPARSE_SOLVE_POLICY_NO_LEVEL	
1	CUSPARSE_SOLVE_POLICY_USE_LEVEL	HIPSPARSE_SOLVE_POLICY_USE_LEVEL	
enum	cusparseStatus_t	hipsparseMatrixType_t	
0	CUSPARSE_STATUS_SUCCESS	HIPSPARSE_STATUS_SUCCESS	
1	CUSPARSE_STATUS_NOT_INITIALIZED	HIPSPARSE_STATUS_NOT_INITIALIZED	
2	CUSPARSE_STATUS_ALLOC_FAILED	HIPSPARSE_STATUS_ALLOC_FAILED	
3	CUSPARSE_STA	HIPSPARSE_STA	

	TUS_INVALID_VALUE	TUS_INVALID_VALUE	
4	CUSPARSE_STATUS_ARCH_MISMATCH	HIPSPARSE_STATUS_ARCH_MISMATCH	
5	CUSPARSE_STATUS_MAPPING_ERROR	HIPSPARSE_STATUS_MAPPING_ERROR	
6	CUSPARSE_STATUS_EXECUTION_FAILED	HIPSPARSE_STATUS_EXECUTION_FAILED	
7	CUSPARSE_STATUS_INTERNAL_ERROR	HIPSPARSE_STATUS_INTERNAL_ERROR	
8	CUSPARSE_STATUS_MATRIX_TYPE_NOT_SUPPORTED	HIPSPARSE_STATUS_MATRIX_TYPE_NOT_SUPPORTED	
9	CUSPARSE_STATUS_ZERO_PIVOT	HIPSPARSE_STATUS_ZERO_PIVOT	
struct	cusparseContext		
typedef	cusparseHandle_t	hipsparseHandle_t	
struct	cusparseHybMat		
typedef	cusparseHybMat_t	hipsparseHybMat_t	
struct	cusparseMatDescr		
typedef	cusparseMatDescr_t	hipsparseMatDescr_t	

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struct	cusparseSolveAnalysisInfo		
typedef	cusparseSolveAnalysisInfo_t		
struct	csrsv2Info		
typedef	csrsv2Info_t	csrsv2Info_t	
struct	csrsm2Info	csrsm2Info	9.2
typedef	csrsm2Info_t	csrsm2Info_t	
struct	bsrsv2Info		
typedef	bsrsv2Info_t		
struct	bsrsm2Info		
typedef	bsrsm2Info_t		
struct	bsric02Info		
typedef	bsric02Info_t		
struct	csrilu02Info		
typedef	csrilu02Info_t	csrilu02Info_t	
struct	bsrilu02Info		
typedef	bsrilu02Info_t		
struct	csru2csrInfo		
typedef	csru2csrInfo_t		
struct	csrgemm2Info	csrgemm2Info	
typedef	csrgemm2Info_t	csrgemm2Info_t	
struct	cusparseColorInfo		
typedef	cusparseColorInfo_t		
struct	pruneInfo		9.0
typedef	pruneInfo_t		9.0
enum	cusparseCsr2Csc		10.1

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	Alg_t		
1	CUSPARSE_CSR2CSC_ALG1		10.1
2	CUSPARSE_CSR2CSC_ALG2		10.1
enum	cusparseFormat_t		10.1
1	CUSPARSE_FORMAT_CSR		10.1
2	CUSPARSE_FORMAT_CSC		10.1
3	CUSPARSE_FORMAT_COO		10.1
4	CUSPARSE_FORMAT_COO_AOS		10.1
enum	cusparseOrder_t		10.1
1	CUSPARSE_ORDER_COL		10.1
2	CUSPARSE_ORDER_ROW		10.1
enum	cusparseSpMVAlg_t		10.1
0	CUSPARSE_MV_ALG_DEFAULT		10.1
1	CUSPARSE_COOMV_ALG		10.1
2	CUSPARSE_CSRMV_ALG1		10.1
3	CUSPARSE_CSRMV_ALG2		10.1

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enum	cusparseSpMMAlg_t		10.1
0	CUSPARSE_MM_ALG_DEFAULT		10.1
1	CUSPARSE_COOMM_ALG1		10.1
2	CUSPARSE_COOMM_ALG2		10.1
3	CUSPARSE_COOMM_ALG3		10.1
4	CUSPARSE_CSRMM_ALG1		10.1
enum	cusparseIndexType_t		10.1
1	CUSPARSE_INDEX_16U		10.1
2	CUSPARSE_INDEX_32I		10.1
3	CUSPARSE_INDEX_64I		10.1
struct	cusparseSpMatDescr		10.1
typedef	cusparseSpMatDescr_t		10.1
struct	cusparseDnMatDescr		10.1
typedef	cusparseDnMatDescr_t		10.1
struct	cusparseSpVecDe		10.1

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	scr		
typedef	cusparseSpVecDescr_t		10.1
struct	cusparseDnVecDescr		10.1
typedef	cusparseDnVecDescr_t		10.1

5.6.2 rocSPARSE 辅助函数

CUDA	HIP	CUDAversion *
cusparseCreate	hipsparseCreate	
cusparseCreateSolveAnalysisInfo		
cusparseCreateHybMat	hipsparseCreateHybMat	
cusparseCreateMatDescr	hipsparseCreateMatDescr	
cusparseDestroy	hipsparseDestroy	
cusparseDestroySolveAnalysisInfo		
cusparseDestroyHybMat	hipsparseDestroyHybMat	
cusparseDestroyMatDescr	hipsparseDestroyMatDescr	
cusparseGetLevelInfo		
cusparseGetMatDiagType	hipsparseGetMatDiagType	
cusparseGetMatFillMode	hipsparseGetMatFillMode	
cusparseGetMatIndexBase	hipsparseGetMatIndexBase	
cusparseGetMatType	hipsparseGetMatType	
cusparseGetPointerMode	hipsparseGetPointerMode	
cusparseGetVersion	hipsparseGetVersion	
cusparseSetMatDiagType	hipsparseSetMatDiagType	

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cusparseSetMatFillMode	hipsparseSetMatFillMode	
cusparseSetMatType	hipsparseSetMatType	
cusparseSetPointerMode	hipsparseSetPointerMode	
cusparseSetStream	hipsparseSetStream	
cusparseGetStream	hipsparseGetStream	8
cusparseCreateCsrsv2Info	hipsparseCreateCsrsv2Info	
cusparseDestroyCsrsv2Info	hipsparseDestroyCsrsv2Info	
cusparseCreateCsrsm2Info	hipsparseCreateCsrsm2Info	9.2
cusparseDestroyCsrsm2Info	hipsparseDestroyCsrsm2Info	9.2
cusparseCreateCsric02Info		
cusparseDestroyCsric02Info		
cusparseCreateCsrilu02Info	hipsparseCreateCsrilu02Info	
cusparseDestroyCsrilu02Info	hipsparseDestroyCsrilu02Info	
cusparseCreateBsrsv2Info		
cusparseDestroyBsrsv2Info		
cusparseCreateBsrsm2Info		
cusparseDestroyBsrsm2Info		
cusparseCreateBsric02Info		
cusparseDestroyBsric02Info		
cusparseCreateBsrilu02Info		
cusparseDestroyBsrilu02Info		
cusparseCreateCsrgemm2Info	hipsparseCreateCsrgemm2Info	
cusparseDestroyCsrgemm2Info	hipsparseDestroyCsrgemm2Info	
cusparseCreatePruneInfo		9

5.6.3 Level1 支持函数列表

CUDA	HIP	CUDAversion
cusparseSaxpyi	hipsparseSaxpyi	

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cusparseDaxpyi	hipsparseDaxpyi	
cusparseCaxpyi	hipsparseCaxpyi	
cusparseZaxpyi	hipsparseZaxpyi	
cusparseSdoti	hipsparseSdoti	
cusparseDdoti	hipsparseDdoti	
cusparseCdoti	hipsparseCdoti	
cusparseZdoti	hipsparseZdoti	
cusparseCdotci	hipsparseCdotci	
cusparseZdotci	hipsparseZdotci	
cusparseSgthr	hipsparseSgthr	
cusparseDgthr	hipsparseDgthr	
cusparseCgthr	hipsparseCgthr	
cusparseZgthr	hipsparseZgthr	
cusparseSgthrz	hipsparseSgthrz	
cusparseDgthrz	hipsparseDgthrz	
cusparseCgthrz	hipsparseCgthrz	
cusparseZgthrz	hipsparseZgthrz	
cusparseSroti	hipsparseSroti	
cusparseDroti	hipsparseDroti	
cusparseSsctr	hipsparseSsctr	
cusparseDsctr	hipsparseDsctr	
cusparseCsctr	hipsparseCsctr	
cusparseZsctr	hipsparseZsctr	

5.6.4 Level2 支持函数列表

CUDA	HIP	CUDA version*
cusparseSbsrmv	hipsparseSbsrmv	
cusparseDbsrmv	hipsparseDbsrmv	

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cusparseCbsrmv	hipsparseCbsrmv	
cusparseZbsrmv	hipsparseZbsrmv	
cusparseSbsrxmv		
cusparseDbsrxmv		
cusparseCbsrxmv		
cusparseZbsrxmv		
cusparseScsrmv	hipsparseScsrmv	
cusparseDcsrmv	hipsparseDcsrmv	
cusparseCcsrmv	hipsparseCcsrmv	
cusparseZcsrmv	hipsparseZcsrmv	
cusparseCsrmvEx		8
cusparseCsrmvEx_bufferSize		8
cusparseScsrmv_mp		8
cusparseDcsrmv_mp		8
cusparseCcsrmv_mp		8
cusparseZcsrmv_mp		8
cusparseSgemvi		7.5
cusparseDgemvi		7.5
cusparseCgemvi		7.5
cusparseZgemvi		7.5
cusparseSgemvi_bufferSize		7.5
cusparseDgemvi_bufferSize		7.5
cusparseCgemvi_bufferSize		7.5
cusparseZgemvi_bufferSize		7.5
cusparseSbsrsv2_bufferSize		
cusparseSbsrsv2_bufferSizeExt		
cusparseDbsrsv2_bufferSize		
cusparseDbsrsv2_bufferSizeExt		

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cusparseCbsrsv2_bufferSize		
cusparseCbsrsv2_bufferSizeExt		
cusparseZbsrsv2_bufferSize		
cusparseZbsrsv2_bufferSizeExt		
cusparseSbsrsv2_analysis		
cusparseDbsrsv2_analysis		
cusparseCbsrsv2_analysis		
cusparseZbsrsv2_analysis		
cusparseXbsrsv2_zeroPivot		
cusparseSbsrsv2_solve		
cusparseDbsrsv2_solve		
cusparseCbsrsv2_solve		
cusparseZbsrsv2_solve		
cusparseScsrsv_analysis		
cusparseDcsrsv_analysis		
cusparseCcsrsv_analysis		
cusparseZcsrsv_analysis		
cusparseCsrsv_analysisEx		8
cusparseScsrsv_solve		
cusparseDcsrsv_solve		
cusparseCcsrsv_solve		
cusparseZcsrsv_solve		
cusparseCsrsv_solveEx		8
cusparseScsrsv2_bufferSize	hipsparseScsrsv2_bufferSize	
cusparseScsrsv2_bufferSizeExt	hipsparseScsrsv2_bufferSizeExt	
cusparseDcsrsv2_bufferSize	hipsparseDcsrsv2_bufferSize	
cusparseDcsrsv2_bufferSizeExt	hipsparseDcsrsv2_bufferSizeExt	
cusparseCcsrsv2_bufferSize	hipsparseCcsrsv2_bufferSize	

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cusparseCcsrsv2_bufferSizeExt	hipsparseCcsrsv2_bufferSizeExt	
cusparseZcsrsv2_bufferSize	hipsparseZcsrsv2_bufferSize	
cusparseZcsrsv2_bufferSizeExt	hipsparseZcsrsv2_bufferSizeExt	
cusparseScsrsv2_analysis	hipsparseScsrsv2_analysis	
cusparseDcsrsv2_analysis	hipsparseDcsrsv2_analysis	
cusparseCcsrsv2_analysis	hipsparseCcsrsv2_analysis	
cusparseZcsrsv2_analysis	hipsparseZcsrsv2_analysis	
cusparseScsrsv2_solve	hipsparseScsrsv2_solve	
cusparseDcsrsv2_solve	hipsparseDcsrsv2_solve	
cusparseCcsrsv2_solve	hipsparseCcsrsv2_solve	
cusparseZcsrsv2_solve	hipsparseZcsrsv2_solve	
cusparseXcsrsv2_zeroPivot	hipsparseXcsrsv2_zeroPivot	
cusparseShybmv	hipsparseShybmv	
cusparseDhybmv	hipsparseDhybmv	
cusparseChybmv	hipsparseChybmv	
cusparseZhybmv	hipsparseZhybmv	
cusparseShybsv_analysis		
cusparseDhybsv_analysis		
cusparseChybsv_analysis		
cusparseZhybsv_analysis		
cusparseShybsv_solve		
cusparseDhybsv_solve		

5.6.5 Level3 支持函数列表

CUDA	HIP	CUDA version*
cusparseDcsrmm	hipsparseDcsrmm	
cusparseCcsrmm	hipsparseCcsrmm	
cusparseZcsrmm	hipsparseZcsrmm	

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cusparseScsrmm2	hipsparseScsrmm2	
cusparseDcsrmm2	hipsparseDcsrmm2	
cusparseCcsrmm2	hipsparseCcsrmm2	
cusparseZcsrmm2	hipsparseZcsrmm2	
cusparseScsrsm_analysis		
cusparseDcsrsm_analysis		
cusparseCcsrsm_analysis		
cusparseZcsrsm_analysis		
cusparseScsrsm_solve		
cusparseDcsrsm_solve		
cusparseCcsrsm_solve		
cusparseZcsrsm_solve		
cusparseScsrsm2_bufferSizeExt	hipsparseScsrsm2_bufferSizeExt	9.2
cusparseDcsrsm2_bufferSizeExt	hipsparseDcsrsm2_bufferSizeExt	9.2
cusparseCcsrsm2_bufferSizeExt	hipsparseCcsrsm2_bufferSizeExt	9.2
cusparseZcsrsm2_bufferSizeExt	hipsparseZcsrsm2_bufferSizeExt	9.2
cusparseScsrsm2_analysis	hipsparseScsrsm2_analysis	9.2
cusparseDcsrsm2_analysis	hipsparseDcsrsm2_analysis	9.2
cusparseCcsrsm2_analysis	hipsparseCcsrsm2_analysis	9.2
cusparseZcsrsm2_analysis	hipsparseZcsrsm2_analysis	9.2
cusparseScsrsm2_solve	hipsparseScsrsm2_solve	9.2
cusparseDcsrsm2_solve	hipsparseDcsrsm2_solve	9.2
cusparseCcsrsm2_solve	hipsparseCcsrsm2_solve	9.2
cusparseZcsrsm2_solve	hipsparseZcsrsm2_solve	9.2
cusparseXcsrsm2_zeroPivot	hipsparseXcsrsm2_zeroPivot	9.2
cusparseSbsrmm		
cusparseDbsrmm		
cusparseCbsrmm		

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cusparseZbsrmm		
cusparseSbsrsm2_bufferSize		
cusparseSbsrsm2_bufferSizeExt		
cusparseDbsrsm2_bufferSize		
cusparseDbsrsm2_bufferSizeExt		
cusparseCbsrsm2_bufferSize		
cusparseCbsrsm2_bufferSizeExt		
cusparseZbsrsm2_bufferSize		
cusparseZbsrsm2_bufferSizeExt		
cusparseSbsrsm2_analysis		
cusparseDbsrsm2_analysis		
cusparseCbsrsm2_analysis		
cusparseZbsrsm2_analysis		
cusparseSbsrsm2_solve		
cusparseDbsrsm2_solve		
cusparseCbsrsm2_solve		
cusparseZbsrsm2_solve		
cusparseXbsrsm2_zeroPivot		
cusparseSgemmi		8
cusparseDgemmi		8
cusparseCgemmi		8
cusparseZgemmi		8

5.6.6 其他函数列表

CUDA	HIP	CUDA version *
cusparseXcsrgeamNnz	hipsparseXcsrgeamNnz	
cusparseScsrgeam	hipsparseScsrgeam	

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cusparseDcsrgeam	hipsparseDcsrgeam	
cusparseCcsrgeam	hipsparseCcsrgeam	
cusparseZcsrgeam	hipsparseZcsrgeam	
cusparseXcsrgeam2Nnz	hipsparseXcsrgeam2Nnz	9.2
cusparseScsrgeam2	hipsparseScsrgeam2	9.2
cusparseDcsrgeam2	hipsparseDcsrgeam2	9.2
cusparseCcsrgeam2	hipsparseCcsrgeam2	9.2
cusparseZcsrgeam2	hipsparseZcsrgeam2	9.2
cusparseScsrgeam2_bufferSizeExt	hipsparseScsrgeam2_bufferSizeExt	9.2
cusparseDcsrgeam2_bufferSizeExt	hipsparseDcsrgeam2_bufferSizeExt	9.2
cusparseCcsrgeam2_bufferSizeExt	hipsparseCcsrgeam2_bufferSizeExt	9.2
cusparseZcsrgeam2_bufferSizeExt	hipsparseZcsrgeam2_bufferSizeExt	9.2
cusparseXcsrgemmNnz	hipsparseXcsrgemmNnz	
cusparseScsrgemm	hipsparseScsrgemm	
cusparseDcsrgemm	hipsparseDcsrgemm	
cusparseCcsrgemm	hipsparseCcsrgemm	
cusparseZcsrgemm	hipsparseZcsrgemm	
cusparseXcsrgemm2Nnz	hipsparseXcsrgemm2Nnz	
cusparseScsrgemm2	hipsparseScsrgemm2	
cusparseDcsrgemm2	hipsparseDcsrgemm2	
cusparseCcsrgemm2	hipsparseCcsrgemm2	
cusparseZcsrgemm2	hipsparseZcsrgemm2	
cusparseScsrgemm2_bufferSizeExt	hipsparseScsrgemm2_bufferSizeExt	
cusparseDcsrgemm2_bufferSizeExt	hipsparseDcsrgemm2_bufferSizeExt	
cusparseCcsrgemm2_bufferSizeExt	hipsparseCcsrgemm2_bufferSizeExt	

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cusparseZcsrgemm2_bufferSizeExt	hipsparseZcsrgemm2_bufferSizeExt	

5.6.7 预设条件函数接口

乔里斯基分解

CUDA	HIP	CUDA version*
cusparseScsric0		
cusparseDcsric0		
cusparseCcsric0		
cusparseZcsric0		
cusparseScsric02_bufferSize		
cusparseScsric02_bufferSizeExt		
cusparseDcsric02_bufferSize		
cusparseDcsric02_bufferSizeExt		
cusparseCcsric02_bufferSize		
cusparseCcsric02_bufferSizeExt		
cusparseZcsric02_bufferSize		
cusparseZcsric02_bufferSizeExt		
cusparseScsric02_analysis		
cusparseDcsric02_analysis		
cusparseCcsric02_analysis		
cusparseZcsric02_analysis		
cusparseScsric02		
cusparseDcsric02		
cusparseCcsric02		
cusparseZcsric02		
cusparseXcsric02_zeroPivot		
cusparseSbsric02_bufferSize		
cusparseSbsric02_bufferSizeExt		

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cusparseDbsric02_bufferSize		
cusparseDbsric02_bufferSizeExt		
cusparseCbsric02_bufferSize		
cusparseCbsric02_bufferSizeExt		
cusparseZbsric02_bufferSize		
cusparseZbsric02_bufferSizeExt		
cusparseSbsric02_analysis		
cusparseDbsric02_analysis		
cusparseCbsric02_analysis		
cusparseZbsric02_analysis		
cusparseSbsric02		
cusparseDbsric02		
cusparseCbsric02		
cusparseZbsric02		
cusparseXbsric02_zeroPivot		

LU 分解

CUDA	HIP	CUDA version*
cusparseScsrilu0		
cusparseDcsrilu0		
cusparseCcsrilu0		
cusparseZcsrilu0		
cusparseCsrilu0Ex		8
cusparseScsrilu02_numericBoost		
cusparseDcsrilu02_numericBoost		
cusparseCcsrilu02_numericBoost		
cusparseZcsrilu02_numericBoost		
cusparseXcsrilu02_zeroPivot	hipsparseXcsrilu02_zeroPivot	

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cusparseScsrilu02_bufferSize	hipsparseScsrilu02_bufferSize	
cusparseScsrilu02_bufferSizeExt	hipsparseScsrilu02_bufferSizeExt	
cusparseDcsrilu02_bufferSize	hipsparseDcsrilu02_bufferSize	
cusparseDcsrilu02_bufferSizeExt	hipsparseDcsrilu02_bufferSizeExt	
cusparseCcsrilu02_bufferSize	hipsparseCcsrilu02_bufferSize	
cusparseCcsrilu02_bufferSizeExt	hipsparseCcsrilu02_bufferSizeExt	
cusparseZcsrilu02_bufferSize	hipsparseZcsrilu02_bufferSize	
cusparseZcsrilu02_bufferSizeExt	hipsparseZcsrilu02_bufferSizeExt	
cusparseScsrilu02_analysis	hipsparseScsrilu02_analysis	
cusparseDcsrilu02_analysis	hipsparseDcsrilu02_analysis	
cusparseCcsrilu02_analysis	hipsparseCcsrilu02_analysis	
cusparseZcsrilu02_analysis	hipsparseZcsrilu02_analysis	
cusparseScsrilu02	hipsparseScsrilu02	
cusparseDcsrilu02	hipsparseDcsrilu02	
cusparseCcsrilu02	hipsparseCcsrilu02	
cusparseZcsrilu02	hipsparseZcsrilu02	
cusparseXbsric02_zeroPivot	hipsparseXcsrilu02_zeroPivot	
cusparseSbsrilu02_numericBoost		
cusparseDbsrilu02_numericBoost		
cusparseCbsrilu02_numericBoost		
cusparseZbsrilu02_numericBoost		
cusparseSbsrilu02_bufferSize		
cusparseSbsrilu02_bufferSizeExt		
cusparseDbsrilu02_bufferSize		
cusparseDbsrilu02_bufferSizeExt		
cusparseCbsrilu02_bufferSize		
cusparseCbsrilu02_bufferSizeExt		
cusparseZbsrilu02_bufferSize		

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cusparseZbsrilu02_bufferSizeExt		
cusparseSbsrilu02_analysis		
cusparseDbsrilu02_analysis		
cusparseCbsrilu02_analysis		
cusparseZbsrilu02_analysis		
cusparseSbsrilu02		
cusparseDbsrilu02		
cusparseCbsrilu02		
cusparseZbsrilu02		
cusparseXbsrilu02_zeroPivot		

三对角矩阵求解

CUDA	HIP	CUDA version*
cusparseSgtsv		
cusparseDgtsv		
cusparseCgtsv		
cusparseZgtsv		
cusparseSgtsv_nopivot		
cusparseDgtsv_nopivot		
cusparseCgtsv_nopivot		
cusparseZgtsv_nopivot		
cusparseSgtsv2_bufferSizeExt		9
cusparseDgtsv2_bufferSizeExt		9
cusparseCgtsv2_bufferSizeExt		9
cusparseZgtsv2_bufferSizeExt		9
cusparseSgtsv2		9
cusparseDgtsv2		9
cusparseCgtsv2		9
cusparseZgtsv2		9

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cusparseSgtsv2_nopivot_bufferSizeExt		9
cusparseDgtsv2_nopivot_bufferSizeExt		9
cusparseCgtsv2_nopivot_bufferSizeExt		9
cusparseZgtsv2_nopivot_bufferSizeExt		9
cusparseSgtsv2_nopivot		9
cusparseDgtsv2_nopivot		9
cusparseCgtsv2_nopivot		9
cusparseZgtsv2_nopivot		9

CUDA	HIP	CUDA version*
cusparseSgtsvStridedBatch		
cusparseDgtsvStridedBatch		
cusparseCgtsvStridedBatch		
cusparseZgtsvStridedBatch		
cusparseSgtsv2StridedBatch_bufferSizeExt		9
cusparseDgtsv2StridedBatch_bufferSizeExt		9
cusparseCgtsv2StridedBatch_bufferSizeExt		9
cusparseZgtsv2StridedBatch_bufferSizeExt		9
cusparseSgtsv2StridedBatch		9
cusparseDgtsv2StridedBatch		9
cusparseCgtsv2StridedBatch		9
cusparseZgtsv2StridedBatch		9
cusparseSgtsvInterleavedBatch_bufferSizeExt		9.2
cusparseDgtsvInterleavedBatch_bufferSizeExt		9.2
cusparseCgtsvInterleavedBatch_bufferSizeExt		9.2
cusparseZgtsvInterleavedBatch_bufferSizeExt		9.2
cusparseSgtsvInterleavedBatch		9.2
cusparseDgtsvInterleavedBatch		9.2

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cusparseCgtsvInterleavedBatch		9.2
cusparseZgtsvInterleavedBatch		9.2

PentadiagonalSolve

CUDA	HIP	CUDA version*
cusparseSgpsvInterleavedBatch_bufferSizeExt		9.2
cusparseDgpsvInterleavedBatch_bufferSizeExt		9.2
cusparseCgpsvInterleavedBatch_bufferSizeExt		9.2
cusparseZgpsvInterleavedBatch_bufferSizeExt		9.2
cusparseSgpsvInterleavedBatch		9.2
cusparseDgpsvInterleavedBatch		9.2
cusparseCgpsvInterleavedBatch		9.2
cusparseZgpsvInterleavedBatch		9.2

矩阵重排

CUDA	HIP	CUDA version*
cusparseScsrcolor		9.2
cusparseDcsrcolor		9.2
cusparseCcsrcolor		9.2

格式转换

CUDA	HIP	CUDA version*
cusparseSbsr2csr		
cusparseDbsr2csr		
cusparseCbsr2csr		
cusparseZbsr2csr		
cusparseSgebsr2gebsc_bufferSize		
cusparseSgebsr2gebsc_bufferSizeExt		
cusparseDgebsr2gebsc_bufferSize		
cusparseDgebsr2gebsc_bufferSizeExt		

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cusparseCgebsr2gebsc_bufferSize		
cusparseCgebsr2gebsc_bufferSizeExt		
cusparseZgebsr2gebsc_bufferSize		
cusparseZgebsr2gebsc_bufferSizeExt		
cusparseSgebsr2gebsc		
cusparseDgebsr2gebsc		
cusparseCgebsr2gebsc		
cusparseZgebsr2gebsc		
cusparseSgebsr2gebsr_bufferSize		
cusparseSgebsr2gebsr_bufferSizeExt		
cusparseDgebsr2gebsr_bufferSize		
cusparseDgebsr2gebsr_bufferSizeExt		
cusparseCgebsr2gebsr_bufferSize		
cusparseCgebsr2gebsr_bufferSizeExt		
cusparseZgebsr2gebsr_bufferSize		
cusparseZgebsr2gebsr_bufferSizeExt		
cusparseXgebsr2gebsrNnz		
cusparseSgebsr2gebsr		
cusparseDgebsr2gebsr		
cusparseCgebsr2gebsr		
cusparseZgebsr2gebsr		
cusparseXgebsr2csr		
cusparseSgebsr2csr		
cusparseDgebsr2csr		
cusparseCgebsr2csr		
cusparseZgebsr2csr		
cusparseScsr2gebsr_bufferSize		
cusparseScsr2gebsr_bufferSizeExt		

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cusparseDcsr2gebsr_bufferSize		
cusparseDcsr2gebsr_bufferSizeExt		
cusparseCcsr2gebsr_bufferSize		
cusparseCcsr2gebsr_bufferSizeExt		
cusparseZcsr2gebsr_bufferSize		
cusparseZcsr2gebsr_bufferSizeExt		
cusparseXcsr2gebsrNnz		
cusparseScsr2gebsr		
cusparseDcsr2gebsr		
cusparseCcsr2gebsr		
cusparseZcsr2gebsr		
cusparseXcoo2csr	hipsparseXcoo2csr	
cusparseScsc2dense		
cusparseDcsc2dense		
cusparseCcsc2dense		
cusparseZcsc2dense		
cusparseScsc2hyb		
cusparseDcsc2hyb		
cusparseCcsc2hyb		
cusparseZcsc2hyb		
cusparseXcsr2bsrNnz		
cusparseScsr2bsr		
cusparseDcsr2bsr		
cusparseCcsr2bsr		
cusparseZcsr2bsr		
cusparseXcsr2coo	hipsparseXcsr2coo	
cusparseScsr2csc	hipsparseScsr2csc	
cusparseDcsr2csc	hipsparseDcsr2csc	

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cusparseCcsr2csc	hipsparseCcsr2csc	
cusparseZcsr2csc	hipsparseZcsr2csc	
cusparseCsr2cscEx		8
cusparseCsr2cscEx2		10.1
cusparseCsr2cscEx2_bufferSize		10.1
cusparseScsr2dense		
cusparseDcsr2dense		
cusparseCcsr2dense		
cusparseZcsr2dense		
cusparseScsr2csr_compress		8
cusparseDcsr2csr_compress		8
cusparseCcsr2csr_compress		8
cusparseZcsr2csr_compress		8
cusparseScsr2hyb	hipsparseScsr2hyb	
cusparseDcsr2hyb	hipsparseDcsr2hyb	
cusparseCcsr2hyb	hipsparseCcsr2hyb	
cusparseZcsr2hyb	hipsparseZcsr2hyb	
cusparseSdense2csc		
cusparseDdense2csc		
cusparseCdense2csc		
cusparseZdense2csc		
cusparseSdense2csr		
cusparseDdense2csr		
cusparseCdense2csr		
cusparseZdense2csr		
cusparseSdense2hyb		
cusparseDdense2hyb		
cusparseCdense2hyb		

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cusparseZdense2hyb		
cusparseShyb2csc		
cusparseDhyb2csc		
cusparseChyb2csc		
cusparseZhyb2csc		
cusparseShyb2csr		
cusparseDhyb2csr		
cusparseChyb2csr		
cusparseZhyb2csr		
cusparseShyb2dense		
cusparseDhyb2dense		
cusparseChyb2dense		
cusparseZhyb2dense		
cusparseSnnz	cusparseSnnz	
cusparseDnnz	cusparseDnnz	
cusparseCnnz	cusparseCnnz	
cusparseZnnz	cusparseZnnz	
cusparseCreateIdentityPermutation	hipsparseCreateIdentityPermutation	
cusparseXcoosort_bufferSizeExt	hipsparseXcoosort_bufferSizeExt	
cusparseXcoosortByRow	hipsparseXcoosortByRow	
cusparseXcoosortByColumn	hipsparseXcoosortByColumn	
cusparseXcsrsort_bufferSizeExt	hipsparseXcsrsort_bufferSizeExt	
cusparseXcsrsort	hipsparseXcsrsort	

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cusparseXcscsort_bufferSizeExt	hipsparseXcscsort_bufferSizeExt	
cusparseXcscsort	hipsparseXcscsort	
cusparseCreateCsru2csrInfo		
cusparseDestroyCsru2csrInfo		
cusparseScsru2csr_bufferSizeExt		
cusparseDcsru2csr_bufferSizeExt		
cusparseCcsru2csr_bufferSizeExt		
cusparseZcsru2csr_bufferSizeExt		
cusparseScsru2csr		
cusparseDcsru2csr		
cusparseCcsru2csr		
cusparseZcsru2csr		
cusparseScsr2csru		
cusparseDcsr2csru		
cusparseCcsr2csru		
cusparseZcsr2csru		
cusparseHpruneDense2csr		9
cusparseSpruneDense2csr		9
cusparseDpruneDense2csr		9
cusparseHpruneDense2csr_bufferSizeExt		9
cusparseSpruneDense2csr_bufferSizeExt		9
cusparseDpruneDense2csr_bufferSizeExt		9
cusparseHpruneDense2csrNnz		9
cusparseSpruneDense2csrNnz		9
cusparseDpruneDense2csrNnz		9
cusparseHpruneCsr2csr		9
cusparseSpruneCsr2csr		9

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cusparseDpruneCsr2csr		9
cusparseHpruneCsr2csr_bufferSizeExt		9
cusparseSpruneCsr2csr_bufferSizeExt		9
cusparseDpruneCsr2csr_bufferSizeExt		9
cusparseHpruneCsr2csrNnz		9
cusparseSpruneCsr2csrNnz		9
cusparseDpruneCsr2csrNnz		9
cusparseHpruneDense2csrByPercentage		9
cusparseSpruneDense2csrByPercentage		9
cusparseDpruneDense2csrByPercentage		9
cusparseHpruneDense2csrByPercentage_bufferSizeExt		9
cusparseSpruneDense2csrByPercentage_bufferSizeExt		9
cusparseDpruneDense2csrByPercentage_bufferSizeExt		9
cusparseHpruneDense2csrNnzByPercentage		9
cusparseSpruneDense2csrNnzByPercentage		9
cusparseDpruneDense2csrNnzByPercentage		9
cusparseHpruneCsr2csrByPercentage		9
cusparseSpruneCsr2csrByPercentage		9
cusparseDpruneCsr2csrByPercentage		9
cusparseHpruneCsr2csrByPercentage_bufferSizeExt		9
cusparseSpruneCsr2csrByPercentage_buff		9

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erSizeExt		
cusparseDpruneCsr2csrByPercentage_bufferSizeExt		9
cusparseHpruneCsr2csrNnzByPercentage		9
cusparseSpruneCsr2csrNnzByPercentage		9
cusparseDpruneCsr2csrNnzByPercentage		9
cusparseSnnz_compress	hipsparseSnnz_compress	8
cusparseDnnz_compress	hipsparseDnnz_compress	8
cusparseCnnz_compress	hipsparseCnnz_compress	8
cusparseZnnz_compress	hipsparseZnnz_compress	8

矩阵生成函数

CUDA	HIP	CUDA version*
cusparseCreateCoo		10.1
cusparseCreateCooAoS		10.1
cusparseCreateCsr		10.1
cusparseDestroySpMat		10.1
cusparseCooGet		10.1
cusparseCooAoSGet		10.1
cusparseCsrGet		10.1
cusparseSpMatGetFormat		10.1
cusparseSpMatGetIndexBase		10.1
cusparseSpMatGetValues		10.1
cusparseSpMatSetValues		10.1
cusparseSpMatGetStridedBatch		10.1

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cusparseSpMatSetStridedBatch		10.1
cusparseSpMatGetNumBatches		10.1
cusparseSpMatSetNumBatches		10.1
cusparseCreateSpVec		10.1
cusparseDestroySpVec		10.1
cusparseSpVecGet		10.1
cusparseSpVecGetIndexBase		10.1
cusparseSpVecGetValues		10.1

稠密矩阵和向量生成函数

CUDA	HIP	CUDA version*
cusparseCreateDnMat		10.1
cusparseDestroyDnMat		10.1
cusparseDnMatGet		10.1
cusparseDnMatGetValues		10.1
cusparseDnMatSetValues		10.1
cusparseDnMatSetStridedBatch		10.1
cusparseDnMatGetStridedBatch		10.1
cusparseCreateDnVec		10.1
cusparseDestroyDnVec		10.1
cusparseDnVecGet		10.1
cusparseDnVecGetValues		10.1
cusparseDnVecSetValues		10.1

SpVV 创建函数

CUDA	HIP	CUDA version*
cusparseSpMM		10.1
cusparseSpMM		10.1

SpMV 创建函数

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CUDA	HIP	CUDA version*
cusparseSpMV		10.1
cusparseSpMV_bufferSize		10.1

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